Prediction-Powered Inference via Calibration

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Abstract

We study semisupervised mean estimation with one labeled sample and one unlabeled sample, where a black-box prediction model is available and may be biased. This setting arises, for example, in modern randomized trials, where machine-learning predictors trained on auxiliary data are used to improve precision, and in recent machine-learning applications with abundant unlabeled data. A standard approach in this setting is augmented inverse-probability weighting (AIPW) (Robins et al., 1994), which leverages the prediction model to improve inference while remaining valid even when that model is inaccurate. In this work, we propose a class of two-step regression-based estimators that first calibrate the prediction model on the labeled sample and then average the calibrated predictions over the pooled covariate sample. The resulting estimator retains a simple imputation-based plug-in form while also admitting an AIPW representation in which calibration removes the augmentation term; in particular, it is a targeted minimum loss estimator (Van Der Laan and Rubin, 2006). We implement calibration using isotonic regression, a lightweight, tuning-free method that learns the best monotone transformation of the prediction score. We show that the resulting estimator is root-n consistent and asymptotically normal under weak conditions. It is statistically efficient among estimators based on the calibrated score, and fully efficient when the calibrated limit equals the true regression function. We also clarify its relationship with recent prediction-powered inference (PPI) methods: PPI is typically less efficient than AIPW, while PPI++ is AIPW with empirical-efficiency maximization (Rubin and van der Laan, 2008) and is first-order equivalent to linear calibration (Mincer and Zarnowitz, 1969) and classical prognostic-score regression adjustment (Hansen, 2008).

1 Introduction

We study semisupervised mean estimation with one labeled dataset $\{(X_i, Y_i)\}_{i=1}^n$ and one unlabeled dataset $\{\widetilde{X}_i\}_{i=1}^N$, where outcomes are observed only for a subset sampled completely at random. This is a classical missing-outcome problem in semiparametric statistics and missing-data theory, for which efficient inference with black-box learners is possible using debiased machine learning methods such as augmented inverse-probability weighting (Robins et al., 1994), targeted minimum loss estimation (Van Der Laan and Rubin, 2006), and double machine learning (Chernozhukov et al., 2018a). Recent machine-learning work has referred to this setting as prediction-powered inference, especially in applications with black-box predictors and abundant unlabeled data (Angelopoulos et al., 2023a,b; Zrnic and Candès, 2024). Closely related problems also arise in flexible covariate adjustment for randomized trials, where machine-learning predictors trained on rich data sources can improve precision in small-sample settings (Hansen, 2008; Moore and van der Laan, 2009a; Lin, 2013; Schuler et al., 2022; Demirel et al., 2024).

A standard approach in this setting is to use the labeled sample to debias a plug-in estimator based on a prediction score. If $m(X)$ is a black-box prediction score, the plug-in estimator averages $m(X)$ over the pooled covariate sample, while the labeled sample is used to estimate the residual correction term $\mathbb{E}[Y - m(X)]$. For example, the augmented inverse probability weighted (AIPW) estimator (Robins et al., 1994) is $\frac{1}{N+n}\{\sum_{i=1}^n m(X_i) + \sum_{i=1}^N m(\widetilde{X}_i)\} + \frac{1}{n}\sum_{i=1}^n\{Y_i - m(X_i)\}$, whereas the PPI estimator (Angelopoulos et al., 2023a), given by $\frac{1}{N}\sum_{i=1}^N m(\widetilde{X}_i) + \frac{1}{n}\sum_{i=1}^n\{Y_i - m(X_i)\}$, is a generally less efficient variant that discards covariate information from the labeled sample. These estimators reduce bias, but they correct only the mean bias of the score.

Calibration provides a stronger and widely studied notion of predictive reliability (Zadrozny and Elkan, 2001, 2002; Niculescu-Mizil and Caruana, 2005). A calibrated score is one whose numerical values accurately reflect the observed outcomes it predicts. Rather than correcting only the average bias $\mathbb{E}[Y - m(X)]$, one can use the labeled sample to construct a calibrated score $m_n^\star(X)$ such that, ideally, $\mathbb{E}[Y \mid m_n^\star(X)] = m_n^\star(X)$. This stronger property immediately implies $\mathbb{E}[m_n^\star(X)] = \mathbb{E}[Y]$, so the resulting plug-in estimator is automatically

calibrated for the mean. This is attractive in practice because a useful prediction model may already be available, while a relatively small labeled sample can often recalibrate its output without retraining the underlying model (Roth, 2022). Such recalibration can be implemented using classical post-processing methods such as Platt scaling (Platt et al., 1999), linear calibration (Mincer and Zarnowitz, 1969), histogram binning (Zadrozny and Elkan, 2001), and isotonic regression (Zadrozny and Elkan, 2002). This is especially relevant for modern machine-learning predictors, such as neural networks, which are often poorly calibrated (Bella et al., 2010; Guo et al., 2017; Wang, 2023).

Our proposal is therefore simple: fit a black-box regression model, calibrate it on the labeled sample, and then average the calibrated predictions over the pooled covariate sample. The resulting estimator remains a "predict, calibrate, and average" procedure, yet our main result shows that it is exactly equal to an AIPW estimator built from the calibrated scores, with calibration eliminating the augmentation term. It is therefore a targeted minimum loss estimator (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011), in which calibration plays the role of the targeting step, and a special case of calibrated debiased machine learning (van der Laan et al., 2024c). Calibration thus gives this simple plug-in rule a debiased interpretation with strong efficiency guarantees.

Contributions. We make three main contributions. First, in Section 3, we introduce a general calibration-based framework for semisupervised mean estimation under the joint two-sample model. The framework accommodates classical post-hoc calibration methods, including linear calibration, histogram binning, and isotonic regression; see, in particular, Sections 3.2 and 3.3. Among these, isotonic calibration is especially appealing because it is lightweight, tuning-free, and learns the best monotone transformation of a prediction score.

Second, in Section 3, we show that calibration enforces empirical score equations that yield an exact AIPW representation of the resulting estimator, placing the method directly within the semiparametric framework for debiased estimation. In Section 3.3, we further study linear calibration and relate PPI and PPI++ to linear regression adjustment with $m(X)$ as a prognostic score, thereby clarifying their connection to classical covariate-adjustment methods in randomized trials and missing-at-random settings.

Third, in Section 4, we establish asymptotic normality, valid inference, and efficiency results for the isotonic-calibrated plug-in estimator under the joint two-sample model. We show that it is efficient in the reduced model defined by the best monotone transformation of the prediction score and is therefore more efficient than AIPW based on the uncalibrated score. When this monotone transformation coincides with the true outcome regression, the estimator is nonparametrically efficient. We review the two-sample semisupervised model and its efficiency theory in Section 2, and we provide synthetic and real-data experiments in Section 5, along with `Python` code in Appendix K.

1.1 Related work

Semiparametric statistics and missing data. This paper builds on foundational ideas from the missing-data and semiparametric literatures. Rubin formalized missing at random as a key identification condition for recovering full-data targets from partially observed outcomes (Rubin, 1976). Semiparametric efficiency theory then characterized efficient influence functions, regular asymptotically linear estimators, and one-step debiased procedures (Levit, 1975; Pfanzagl and Wefelmeyer, 1985; Hasminskii and Ibragimov, 1979; Klaassen, 1987; Bickel et al., 1993). In missing-data and causal-inference problems, these ideas led to influence-function-based estimating equations (Robins et al., 1994, 1995; Robins and Rotnitzky, 1995; van der Laan and Robins, 2003; Tsiatis, 2006) that are often doubly robust (Bang and Robins, 2005). From the perspective of asymptotic efficiency, missing-data models under missingness or coarsening at random are now well understood: the class of regular asymptotically linear estimators can be characterized through their influence functions in these models (Robins et al., 1994, 1995; Robins and Rotnitzky, 1995; van der Laan and Robins, 2003) (see Appendix C for a brief review).

Debiased machine learning and targeted learning. A large body of work combines the semiparametric framework above with machine-learning-based nuisance estimation, often under the label of debiased machine

learning. Flexible learning methods in estimating equations and AIPW-style procedures already appear in van der Laan and Robins (2003). Targeted minimum loss estimation (TMLE) (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011) provides a closely related framework in which an initial nuisance estimator is updated in a parameter-targeted way to debias the resulting plug-in estimator; in the present setting, the main nuisance is the outcome regression (Gruber and Van Der Laan, 2009; Hoffman, 2020; Ross et al., 2025). These approaches are often combined with sample splitting and cross-fitting to accommodate generic machine learning estimators (Schick, 1986; Klaassen, 1987; Zheng and Van Der Laan, 2010; van der Laan and Rose, 2011; Chernozhukov et al., 2018a). Double machine learning emphasizes similar ingredients—orthogonality, cross-fitting, and influence-function expansions—for inference with modern nuisance estimators (Chernozhukov et al., 2018a, 2022), and can be understood within the same semiparametric framework (van der Laan, 2019; Díaz, 2020; Chen et al., 2026). Closely related work also studies balancing-weight estimators (Hainmueller, 2012; Imai and Ratkovic, 2014; Zubizarreta, 2015; Lendle et al., 2015; Chattopadhyay et al., 2020; Hejazi and van der Laan, 2023), which often admit equivalent regression-adjustment or augmented representations (Chattopadhyay and Zubizarreta, 2023; Bruns-Smith et al., 2025; Rotnitzky et al., 2025). In completely missing-at-random settings, including semisupervised learning and randomized controlled trials, such estimators are often robust to model misspecification, allowing flexible black-box methods to be used for nuisance estimation while preserving valid inference (Moore and van der Laan, 2009a,b).

Semisupervised inference. Related work studies semisupervised or surrogate-assisted estimation in settings where the primary outcome is observed only on a limited subset, while auxiliary outcomes or surrogate measurements are available more broadly (Pepe, 1992; Pepe et al., 1994; Chen, 2000; Chen et al., 2008; Cheng et al., 2021; Ji et al., 2025; Kallus and Mao, 2025). Two-sample semisupervised inference is a special case of two-stage sampling designs (Scott and Holt, 1982; Rose and van der Laan, 2011; Hejazi et al., 2021; Qiu et al., 2026) and, more generally, of data fusion, for which a substantial semiparametric efficiency literature has been developed (Li and Luedtke, 2023; Li et al., 2025; Graham et al., 2024; Xu et al., 2025). In this context, more recent papers have used the term *prediction-powered inference* for the semisupervised mean-estimation setting in which a black-box predictor is combined with a small labeled sample and a large unlabeled sample (Angelopoulos et al., 2023a,b; Zrnic and Candès, 2024; Xu et al., 2025; Song et al., 2026; Poulet et al., 2025; Lee and Kim, 2026). Closely related problems have also long been studied in randomized trials, where prognostic scores or other regression adjustments learned from larger historical or auxiliary data sources are used to improve precision in smaller trials (Hansen, 2008; Rosenblum and van der Laan, 2009; Moore and van der Laan, 2009a,b; Moore et al., 2011; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). These methods are special cases of earlier semiparametric and debiased machine-learning ideas from the missing-data literature. For example, PPI++ is an AIPW estimator (Robins et al., 1994) obtained by empirical efficiency maximization (Rubin and van der Laan, 2008) over a scaling parameter, whereas cross-PPI (Zrnic and Candès, 2024) is an AIPW estimator with cross-fitting, which accommodates machine-learning nuisance estimators and relaxes empirical process conditions, as in Schick (1986); Zheng and Van Der Laan (2010); van der Laan and Rose (2011); Chernozhukov et al. (2018a). Its stability-based proof strategy is also similar in spirit to the analysis of Chen et al. (2022). More fundamentally, robustness to black-box prediction models is an immediate consequence of the double robustness of AIPW when the missingness probability is known (Bang and Robins, 2005).

From missing data to causal inference. From the perspective of estimation and efficiency theory, the semisupervised setting with missing-at-random outcomes is statistically equivalent to randomized experiments, controlled trials, and survey-sampling designs once the missingness indicator is identified with the treatment or sampling indicator, so that the corresponding potential outcomes are treated as missing (Rubin, 1978; van der Laan and Robins, 2003; Ding and Li, 2018; Mozer, 2026). Accordingly, tools for covariate adjustment, causal inference, and model-assisted estimation developed in those settings can be imported directly into the semisupervised setting; see, for example, Mozer (2026) for equivalence results relating prediction-powered inference to model-assisted survey-sampling estimators. Moreover, the two-sample setting considered here is statistically equivalent to a one-sample i.i.d. setting (Li and Luedtke, 2023). In particular, the methods we develop also apply to inference on the counterfactual mean $\mathbb{E}[Y(1)]$ in a randomized controlled trial with binary treatment $A_i \in \{0,1\}$, where $\{(X_i, Y_i)\}_{i=1}^n$ are the observed covariates and outcomes for units assigned to treatment $A_i = 1$, so that $Y_i = Y_i(1)$, and the unlabeled dataset $\{\widetilde{X}_i\}_{i=1}^N$ consists of covariate information for units assigned to control $A_i = 0$ (see Appendix B).

Calibration. A final relevant thread concerns calibration. Classical post-processing methods such as Platt scaling (Platt et al., 1999; Cox, 1958), linear calibration (Mincer and Zarnowitz, 1969; Chernozhukov et al., 2018b; Leng and Dimmery, 2021), isotonic regression (Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005), and histogram binning (Zadrozny and Elkan, 2001; Gupta and Ramdas, 2021) improve predictive reliability without retraining the underlying model. The idea of calibration has earlier roots in the forecasting literature (Mincer and Zarnowitz, 1969; Lichtenstein et al., 1977; Vovk, 1992; Vovk et al., 2005; Gneiting et al., 2007; Lambert, 2011); see historical discussion of Lee et al. (2023). More recent work studies calibration under broader losses and prediction tasks (Jung et al., 2021; Noarov and Roth, 2023; van der Laan et al., 2023b; Whitehouse et al., 2024; van der Laan and Alaa, 2025). Calibration has also been used in conformal prediction (van der Laan and Alaa, 2024, 2025), importance-weight stabilization (Gutman et al., 2022; Deshpande and Kuleshov, 2023; Ballinari and Bearth, 2024; van der Laan et al., 2024b), treatment effect estimation (Chernozhukov et al., 2018b; van der Laan et al., 2023b; Whitehouse et al., 2024), and value prediction in offline reinforcement learning (van der Laan and Kallus, 2025). Distribution-free guarantees for histogram binning are developed by Gupta and Ramdas (2021), while van der Laan et al. (2023b); van der Laan and Alaa (2025) establish asymptotic distribution-free guarantees for isotonic calibration. Most closely related for our purposes is recent work connecting calibration to semiparametric inference by formulating calibration through score equations and using it to support debiased estimation of statistical functionals (van der Laan et al., 2024c, 2023a, 2025b). In particular, van der Laan et al. (2024c) studies calibrated DML for doubly robust inference, which in the present setting reduces to calibrating the outcome regression and exploiting the resulting plug-in representation, while van der Laan et al. (2025b) studies calibrated plug-in regression estimators in reinforcement learning. More broadly, van der Laan et al. (2023a) studies plug-in calibration for both regression and treatment effect estimation and shows how calibration can support superefficient inference through dimension reduction.

2 Two-Sample Semisupervised Setup

2.1 Data structure and notation

We observe two independent datasets:

$$\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n \qquad \text{and} \qquad \mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N.$$

The labeled sample is i.i.d. from a distribution P_0 on $\mathcal{X} \times \mathbb{R}$, while the unlabeled sample is i.i.d. from the marginal distribution $P_{0,X}$ of X under P_0. We target the population mean

$$\psi_0 := \mathbb{E}_{P_0}[Y], \qquad \mu_0(x) := \mathbb{E}_{P_0}[Y \mid X = x].$$

It is helpful to keep the joint two-sample experiment explicit. For each pair (n, N), the data law is

$$\mathbb{D}_0^{(n,N)} := P_0^{\otimes n} \otimes P_{0,X}^{\otimes N},$$

and the target parameter is the functional

$$\Psi\left(\mathbb{D}_0^{(n,N)}\right) := \psi_0 = \int y \, dP_0(x, y) = \int \mu_0(x) \, dP_{0,X}(x).$$

For measurable functions $a : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ and $b : \mathcal{X} \to \mathbb{R}$, define the empirical means

$$\mathbb{P}_n^L a := \frac{1}{n} \sum_{i=1}^n a(X_i, Y_i), \qquad \mathbb{P}_N^U b := \frac{1}{N} \sum_{j=1}^N b(\widetilde{X}_j).$$

For clarity, we will sometimes abuse notation and write, for example, $\mathbb{P}_n^L\{a(X, Y)\}$ for $\mathbb{P}_n^L a$ and, when a function depends only on X, write $\mathbb{P}_n^L\{b(X)\} := \frac{1}{n} \sum_{i=1}^n b(X_i)$. Throughout, we write $\rho_n = n/M$ for the

labeled fraction, equivalently the known probability of observing the outcome, and use the identity

$$\frac{1}{n+N}\left\{\sum_{i=1}^{n} f(X_i) + \sum_{j=1}^{N} f(\widetilde{X}_j)\right\} = \rho_n \mathbb{P}_n^L f + (1-\rho_n)\mathbb{P}_N^U f.$$

That is, the empirical mean of $f(X)$ over the pooled covariate sample is a convex combination of the empirical means in the labeled and unlabeled samples.

Assumption 1 (Two-sample design)**.** The following hold:

(i) $\mathcal{D}_L$ is i.i.d. from P_0, and $\mathcal{D}_U$ is i.i.d. from $P_{0,X}$.

(ii) $\mathcal{D}_L$ and $\mathcal{D}_U$ are independent.

(iii) Y has finite second moment under P_0.

(iv) With $M := n + N$, the labeled fraction $\rho_n := n/M$ satisfies $\rho_n \to \rho_0$ for some $\rho_0 \in (0, 1)$.

Remark 1. The labeled sample carries the outcome information, while the unlabeled sample sharpens estimation of the marginal covariate distribution. The pooled i.i.d. missing-data formulation in Appendix I is a special case of this two-sample setup.

2.2 Review of AIPW, PPI, and semiparametric efficiency

For later reference, we briefly review semiparametric efficiency in the unrestricted two-sample model (Bickel et al., 1993; Li and Luedtke, 2023), along with the associated class of influence functions and AIPW estimators (Robins et al., 1994, 1995; Bang and Robins, 2005). We say that an estimator $\widehat{\psi}$ is asymptotically linear with *influence pair* (D^L, D^U) if

$$\sqrt{M}\,(\widehat{\psi} - \psi_0) = \frac{1}{\sqrt{M}}\sum_{i=1}^{n} D^L(X_i, Y_i) + \frac{1}{\sqrt{M}}\sum_{j=1}^{N} D^U(\widetilde{X}_j) + o_p(1),$$

where $\mathbb{E}_{P_0}[D^L(X, Y)] = 0$ and $\mathbb{E}_{P_{0,X}}[D^U(\widetilde{X})] = 0$. In other words, the error $\widehat{\psi} - \psi_0$ is asymptotically equivalent to the empirical mean $\frac{1}{M}\{\sum_{i=1}^{n} D^L(X_i, Y_i) + \sum_{j=1}^{N} D^U(\widetilde{X}_j)\}$. This is the natural two-sample analogue of the usual influence function in an i.i.d. model (Van der Vaart, 2000) and uniquely determines the corresponding influence function in the i.n.i.d. model (Bickel et al., 1993). By the central limit theorem and the independence of the two datasets,

$$\sqrt{M}\,(\widehat{\psi} - \psi_0) \;\rightsquigarrow\; N\left(0,\, \rho_0 \mathrm{Var}_{P_0}\big(D^L(X, Y)\big) + (1-\rho_0)\mathrm{Var}_{P_{0,X}}\big(D^U(\widetilde{X})\big)\right).$$

Thus, the influence pair (D^L, D^U) determines the asymptotic variance of $\widehat{\psi}$.

The following result characterizes the class of regular estimators[1] (Van der Vaart, 2000) in this model through their influence functions and shows that each is asymptotically equivalent to an AIPW estimator. This provides the benchmark class for our later analysis, where we place calibrated plug-in estimators within the standard AIPW framework and characterize their efficiency. The result follows from Robins et al. (1994) together with the statistical equivalence between the two-sample model and the single-sample i.i.d. model established, for example, by Li and Luedtke (2023).

[1]Regularity means the estimators limiting distrivution is invariant under sampling from local alternatives; it excludes pathological estimators, such as Hodge's estimator, that may be superefficient at specific laws but performs poorly in a local asymptotic sense (Le Cam, 1953; Van der Vaart, 2000)

Proposition 1 (AIPW class of influence pairs)**.** *Under Assumption 1, a mean-zero pair (D^L, D^U) is the influence pair of a regular estimator of ψ_0 in the unrestricted two-sample model if and only if there exists a square-integrable function $f : \mathcal{X} \to \mathbb{R}$ such that, with*

$$\tilde{f}(X) := f(X) - \mathbb{E}_{P_{0,X}}[f(X)] + \psi_0,$$

the pair takes the form

$$D_f^L(X, Y) = \tilde{f}(X) - \psi_0 + \rho_0^{-1}\{Y - \tilde{f}(X)\}, \tag{1}$$

$$D_f^U(\widetilde{X}) = \tilde{f}(\widetilde{X}) - \psi_0. \tag{2}$$

Furthermore, for any such $f : \mathcal{X} \to \mathbb{R}$, a corresponding estimator with this influence pair is given by the AIPW estimator

$$\widehat{\psi}(f) := \rho_n \mathbb{P}_n^L\{f(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{f(\widetilde{X})\} + \mathbb{P}_n^L\{Y - f(X)\}. \tag{3}$$

AIPW, PPI, PPI++, and TMLE. Proposition 1 implies that, for any fixed choice of $f : \mathcal{X} \to \mathbb{R}$, the estimator $\widehat{\psi}(f)$ in (3) is unbiased in finite samples and asymptotically linear with the corresponding influence pair. A classical choice is $f(X) = m(X)$, where $m(X)$ denotes the prediction score, which yields the usual AIPW estimator (Robins et al., 1994):

$$\widehat{\psi}_{\text{AIPW}} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

This estimator can also be written in the combined-sample form

$$\widehat{\psi}_{\text{AIPW}} = \frac{1}{M}\left\{\sum_{i=1}^{n} m(X_i) + \sum_{i=1}^{N} m(\widetilde{X}_i) + \frac{1}{\rho_n}\sum_{i=1}^{n}\{Y_i - m(X_i)\}\right\},$$

where ρ_n^{-1} is the known inverse probability of observing the outcome. Because outcome missingness is completely at random with known constant probability ρ_n, the AIPW estimator remains valid even if m is misspecified; the prediction score affects only efficiency. Thus, AIPW is "safe" in this setting (Xu et al., 2025). This is a special case of double robustness (Bang and Robins, 2005).

An alternative choice takes $f(X) = m(X)/(1 - \rho_n)$, which yields

$$\widehat{\psi}_{\text{PPI}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}.$$

This choice is used implicitly in Angelopoulos et al. (2023a), where it is termed PPI[2]. Relative to $\widehat{\psi}_{\text{AIPW}}$, $\widehat{\psi}_{\text{PPI}}$ omits the labeled-sample average of $m(X)$ from the plug-in term. It is therefore typically less efficient, especially when m is reasonably accurate; in particular, it is inefficient when $m = \mu_0$. When the labeled fraction ρ_n is small and m is inaccurate, however, the AIPW estimator need not be more efficient than the PPI estimator (Han, 2012; Rotnitzky et al., 2012), particularly when $\text{Cov}(m(X), Y - m(X))$ is sufficiently positive, so that the cross-correlation between the plug-in term and the residual-correction term increases variance rather than reducing it.

Because both $\widehat{\psi}_{\text{AIPW}}$ and $\widehat{\psi}_{\text{PPI}}$ are special cases of the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, one can construct an estimator that is asymptotically at least as efficient as both by minimizing the empirical influence-function variance over this class:

$$\sigma_n^2(f) := \rho_n \mathbb{P}_n^L\left[\left\{f(X) - \widehat{\psi}(f) + \rho_n^{-1}(Y - f(X))\right\}^2\right] + (1 - \rho_n)\mathbb{P}_N^U\left[\left\{f(\widetilde{X}) - \widehat{\psi}(f)\right\}^2\right].$$

Equivalently, one chooses λ by empirical efficiency maximization over $\{\lambda m(X) : \lambda \in \mathbb{R}\}$ (van der Laan and Robins, 2003; Rubin and van der Laan, 2008); related ideas were used by Rotnitzky et al. (2012) to construct doubly robust estimators with improved efficiency under misspecification. The resulting estimator, called

[2]The PPI and PPI++ estimators are algebraically equivalent to the model-assisted survey-design estimators of Cassel et al. (1976) and Särndal et al. (2003), respectively; see Mozer (2026) for a review.

PPI++ by Angelopoulos et al. (2023b), is simply AIPW with an empirically rescaled prediction score $\hat{\lambda} m(X)$, where $\hat{\lambda}$ estimates the population-optimal scaling coefficient

$$\lambda^\star = \frac{\mathrm{Cov}\big(Y, m(X)\big)}{\mathrm{Var}\big(m(X)\big)}. \tag{4}$$

Efficiency can be improved further by minimizing the empirical efficiency criterion $\sigma_n^2(f)$ over richer classes, such as

$$\{\lambda m(X) + \beta_{\mathrm{lin}}^\top X : \lambda \in \mathbb{R},\ \beta_{\mathrm{lin}} \in \mathbb{R}^d\},$$

or over a reproducing kernel Hilbert space. More generally, the empirical efficiency criterion can be used in a cross-validation procedure to select among different estimator classes (Rubin and van der Laan, 2008).

The semiparametrically efficient benchmark is attained by the unrestricted choice $f(X) = \mu_0(X)$, which must typically be estimated from the data. From the balancing perspective, this shows that efficiency generally requires using the full covariate information X, or at least a summary rich enough to recover the true regression function $\mu_0(X)$ (Robins et al., 1994). The next result identifies the efficient influence pair, and thus the most efficient regular estimator, in this model (Bickel et al., 1993). Accordingly, efficiency is optimized by using a prediction model $m(X)$ that approximates $\mu_0(X)$ as closely as possible (Moore and van der Laan, 2009a,b). Modern debiased machine learning methods implement this principle using flexible black-box learners to estimate nuisance functions (van der Laan and Robins, 2003; Moore and van der Laan, 2009a; van der Laan and Rose, 2011; Chernozhukov et al., 2018a).

Proposition 2 (Efficient influence pair in the full model)**.** *Under Assumption 1, the asymptotic variance over the class equations (1) and (2) is minimized at $f = \mu_0$. The efficient influence pair for $\psi_0 = \mathbb{E}_{P_0}[Y]$ in the unrestricted two-sample model is therefore*

$$D_{\mathrm{eff}}^L(X, Y) = \mu_0(X) - \psi_0 + \rho_0^{-1}\{Y - \mu_0(X)\}, \tag{5}$$

$$D_{\mathrm{eff}}^U(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0. \tag{6}$$

To conclude, we illustrate how regression adjustment, balancing-weight adjustment, and AIPW estimation are closely connected (Rotnitzky et al., 2025). We also introduce TMLE (Van Der Laan and Rubin, 2006), which provides an alternative framework for debiased plug-in estimation without explicit augmentation.

Remark 2 (Plug-in approaches to bias correction and TMLE)**. Regression plug-in estimators.** A particularly simple realization of this first-order class is obtained through plug-in regression adjustment (Scharfstein et al., 1999; Van Der Laan and Rubin, 2006; Rosenblum and van der Laan, 2009; Rotnitzky et al., 2025). Let

$$m_n(X) = \hat{a}_{\mathrm{lin}}\, m(X) + \hat{b}_{\mathrm{lin}}, \qquad (\hat{a}_{\mathrm{lin}}, \hat{b}_{\mathrm{lin}}) = \arg\min_{a,b} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2.$$

The corresponding plug-in estimator is

$$\rho_n \mathbb{P}_n^L\{m_n(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\}.$$

For any $\lambda \in \mathbb{R}$, this estimator also admits the AIPW representation

$$\rho_n \mathbb{P}_n^L\{m_n(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\} + \lambda \mathbb{P}_n^L\big[Y - m_n(X)\big],$$

because the augmentation term $\mathbb{P}_n^L[Y - m_n(X)]$ vanishes by the least-squares normal equations. Thus, when m_n converges to the best linear predictor of Y given $m(X)$, this plug-in estimator is typically at least as efficient as $\widehat{\psi}_{\mathrm{AIPW}}$ and $\widehat{\psi}_{\mathrm{PPI}}$, since it corresponds to an influence function with smaller or equal variance.

Balancing-weight estimators. Dually, the same estimator can be expressed through balancing weights. Let $w_n(X) = \hat{a}_{\mathrm{bal}} + \hat{b}_{\mathrm{bal}}\, m(X)$ be affine weights that balance $(1, m(X))$, in the sense that

$$\mathbb{P}_n^L\{w_n(X)\} = 1 \quad \text{and} \quad \mathbb{P}_n^L\big[m(X) w_n(X)\big] = \rho_n \mathbb{P}_n^L\{m(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\}.$$

Then the corresponding balancing-weight estimator satisfies

$$\mathbb{P}_n^L\{w_n(X)Y\} = \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\big[w_n(X)\{Y - m(X)\}\big].$$

In fact, the plug-in and balancing-weight estimators are algebraically equivalent (Chattopadhyay and Zubizarreta, 2023; Bruns-Smith et al., 2025):

$$\mathbb{P}_n^L\{w_n(X)Y\} = \rho_n \mathbb{P}_n^L\{m_n(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n(\widetilde{X})\}.$$

TMLE. More broadly, such debiased plug-in estimators can be viewed as instances of TMLE (Van Der Laan and Rubin, 2006; van der Laan and Rose, 2011), a general framework for debiased and efficient plug-in estimation that targets nuisance components to eliminate first-order bias. In the present setting, either the weights or the regression model are updated so that the corresponding augmentation term vanishes. In weighting-based TMLE, the update balances on the prediction model $m(X)$ (e.g., Hejazi and van der Laan 2023); one may either include the known weight $1/\rho_n$ as an offset or balance on the intercept as well. In regression-based TMLE, the prediction model $m(X)$ may be included either as a covariate or as an offset (e.g., Lendle et al. 2015), and the intercept adjustment corrects for the known inverse missingness-probability weight $1/\rho_n$. The targeting step is ultimately a practical choice, provided it solves the relevant estimating equation and removes the augmentation term; different choices may also yield stronger properties, such as additional bias reduction or doubly robust inference (Van der Laan, 2014; Carone et al., 2014; Díaz and van der Laan, 2017; Benkeser et al., 2017; van der Laan et al., 2021).

3 Calibration-Based Plug-In Estimation

The first subsection introduces the general calibrated plug-in estimator and shows that it admits an equivalent AIPW representation. The second subsection then describes concrete ways to enforce empirical calibration, focusing on isotonic regression. We next discuss linear calibration as a simpler adjustment step when data are especially scarce. The final subsection presents a cross-fitted version of the same construction, which allows the predictive score to be learned from the labeled data.

3.1 General estimator and exact AIPW representation

Let $m : \mathcal{X} \to \mathbb{R}$ denote a black-box regression fit on the labeled data. A calibration procedure takes the labeled scores $m(X_1), \ldots, m(X_n)$ and outcomes $Y_1, \ldots, Y_n$, fits a post-hoc map f_n, and outputs the calibrated regression

$$m_n^\star := f_n \circ m.$$

Thus, the fitted values on the two samples are $m_n^\star(X_i)$ for $i = 1, \ldots, n$ and $m_n^\star(\widetilde{X}_j)$ for $j = 1, \ldots, N$.

Fix a linear class $\mathcal{F}$ of real-valued functions $f : \mathbb{R} \to \mathbb{R}$ that contains the constant functions $t \mapsto c$ and the identity map $t \mapsto t$. Informally, calibration is a reliability property stating that the prediction model $m_n^\star$ cannot be improved by post-processing its output using transformations in $\mathcal{F}$. We say that $m_n^\star$ is *empirically calibrated over* $\mathcal{F}$ if no admissible transformation of its output improves the empirical squared risk, that is, if for every $f \in \mathcal{F}$,

$$\mathbb{P}_n^L\big[(Y - m_n^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - f(m_n^\star(X)))^2\big], \tag{7}$$

where $\mathbb{P}_n^L$ denotes the empirical mean over the labeled sample. Equivalently, the corresponding first-order optimality conditions imply

$$\mathbb{P}_n^L\big[f(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = 0, \qquad f \in \mathcal{F}. \tag{8}$$

Because $\mathcal{F}$ contains the constant functions and the identity map, empirical calibration implies, in particular, the mean-calibration and linear-calibration identities

$$\mathbb{P}_n^L\{Y - m_n^\star(X)\} = 0, \qquad \mathbb{P}_n^L\big[m_n^\star(X)\{Y - m_n^\star(X)\}\big] = 0.$$

The empirically calibrated plug-in estimator is

$$\widehat{\psi}_{\mathrm{cal}} := \frac{1}{n+N}\left\{\sum_{i=1}^{n} m_n^\star(X_i) + \sum_{j=1}^{N} m_n^\star(\widetilde{X}_j)\right\}. \tag{9}$$

Equivalently,

$$\widehat{\psi}_{\mathrm{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\}.$$

We focus on two cases. In the nonparametric case, $\mathcal{F}$ is the class of all transformations $f:\mathbb{R}\to\mathbb{R}$, which we refer to as *full calibration*; in practice, this can be implemented by regressing Y on $m(X)$ using histogram regression or isotonic regression (Section 3.2). In the parametric case, $\mathcal{F}$ is restricted to affine transformations $t \mapsto at+b$, which we refer to as *linear calibration*; this amounts to fitting the working linear model $a\,m(X)+b$ for Y (Section 3.3). More generally, if $\mathcal{F}$ is finite-dimensional with basis $\{\varphi_1,\ldots,\varphi_p\}$, then an empirically calibrated predictor m_n^* can be constructed by regressing Y on $\{\varphi_1(m(X)),\ldots,\varphi_p(m(X))\}$.[3]

The key property of calibrated plug-in estimators is the following.

Theorem 1 (Exact AIPW representation under empirical calibration)**.** *Suppose the calibrated predictions satisfy equation (8), where $\mathcal{F}$ is a linear class containing the constant functions and the identity map. Then, for any empirical balancing weight $\widehat{w}\in\mathcal{F}$, the calibrated plug-in estimator admits the exact AIPW representation*

$$\widehat{\psi}_{\mathrm{cal}} = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} + \rho_n\mathbb{P}_n^L\big[\widehat{w}(m_n^\star(X))\{Y-m_n^\star(X)\}\big]. \tag{10}$$

Moreover, if $\widehat{w}_n^*\in\mathcal{F}$ balances $\mathcal{F}$ in the sense that

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\,f(m_n^\star(X))\big] = \rho_n\mathbb{P}_n^L\{f(m_n^\star(X))\} + (1-\rho_n)\mathbb{P}_N^U\{f(m_n^\star(\widetilde{X}))\}, \qquad f\in\mathcal{F}, \tag{11}$$

then $\widehat{\psi}_{\mathrm{cal}}$ admits the weighted-outcome representation

$$\widehat{\psi}_{\mathrm{cal}} = \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))Y\big]. \tag{12}$$

Theorem 1 shows that the calibrated plug-in estimator $\widehat{\psi}_{\mathrm{cal}}$ is exactly equal to the AIPW estimator with $m_n^\star$ as the prediction score. It also yields a dual representation of $\widehat{\psi}_{\mathrm{cal}}$ as a balancing-weighted outcome estimator, where $\widehat{w}_n^*$ is chosen to balance moments of functions in $\mathcal{F}$ between the labeled and unlabeled samples (Chattopadhyay et al., 2020). More broadly, $\widehat{\psi}_{\mathrm{cal}}$ may also be viewed as a TMLE, with calibration playing the role of the targeting step.

Sketch of asymptotic theory. We now sketch the asymptotic theory for $\widehat{\psi}_{\mathrm{cal}}$. Suppose the estimated calibrated score $m_n^\star$ converges to a limit m_0, typically a transformation of m that minimizes a population risk over a suitable class. This limit is typically calibrated at the population level in the sense that

$$\mathbb{E}_{P_0}\big[f\{m_0(X)\}\{\mu_0(X)-m_0(X)\}\big] = 0, \qquad f\in\mathcal{F}.$$

If $\mathcal{F}$ contains all real-valued functions, then

$$m_0(X) = \mathbb{E}_{P_0}\left[Y \mid m_0(X)\right],$$

so the predictive performance of m_0 cannot be improved by any transformation of its output. Taking $\hat{w}=1$ in (10), $\widehat{\psi}_{\mathrm{cal}}$ can be viewed as a plug-in version of the usual AIPW estimator with $m_n^\star$ as the regression adjustment. Therefore, if $m_n^\star$ is sufficiently close to m_0, then $\widehat{\psi}_{\mathrm{cal}}$ should behave asymptotically like the corresponding oracle AIPW estimator:

$$\begin{aligned}\widehat{\psi}_{\mathrm{cal}} - \psi_0 &= \rho_n(\mathbb{P}_n^L - P_0)\Big\{m_0(X) - \psi_0 + \rho_n^{-1}\big(Y - m_0(X)\big)\Big\} \\ &\quad + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\big\{m_0(\widetilde{X}) - \psi_0\big\} + o_p(M^{-1/2}).\end{aligned}$$

In particular, $\widehat{\psi}_{\mathrm{cal}}$ is asymptotically linear with influence pair

$$D_{m_0}^L(X,Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \qquad D_{m_0}^U(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0.$$

In Section 4, we establish this result for the isotonic-calibrated plug-in estimator introduced next.

[3] If $\mathcal{F}$ does not contain the identity map, one can instead include $m(X)$ as an offset, and (8) still holds.

Algorithm 1 Isotonic-calibrated plug-in estimator

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $\hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \{Y_i - f(T_i)\}^2$
3: Set $m^\star_{n,\text{iso}}(X_i) = \hat{f}(T_i)$ and $m^\star_{n,\text{iso}}(\widetilde{X}_j) = \hat{f}(\widetilde{T}_j)$
4: Form $\widehat{\psi}_{\text{iso}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\text{iso}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{iso}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{iso}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}}$
6: Set $\widehat{\text{SE}}^2_{\text{iso}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\widehat{\text{SE}}_{\text{iso}}$

3.2 Full calibration via isotonic regression

In practice, full empirical calibration can be achieved by fitting a second-stage calibrator to the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$ (van der Laan and Alaa, 2025). Two especially simple options are histogram binning and isotonic calibration (Zadrozny and Elkan, 2001, 2002; Niculescu-Mizil and Caruana, 2005), both of which yield exact empirical calibration. Histogram binning fits a piecewise-constant map on a prespecified partition of the score (Gupta and Ramdas, 2021), whereas isotonic calibration estimates the best monotone transformation of the score and adaptively determines the effective bins using the pooled-adjacent-violators algorithm (Barlow and Brunk, 1972). We focus on isotonic regression because it is lightweight, tuning-free, and easy to implement using widely available software. Moreover, because the identity map is monotone, isotonic calibration can only improve the empirical mean squared error over $\mathcal{D}_L$ relative to the original scores.

The isotonic-calibrated plug-in estimator is

$$\widehat{\psi}_{\text{iso}} := \frac{1}{n+N}\left\{\sum_{i=1}^n m^\star_{n,\text{iso}}(X_i) + \sum_{j=1}^N m^\star_{n,\text{iso}}(\widetilde{X}_j)\right\},$$

where

$$m^\star_{n,\text{iso}}(X) = \hat{f}\{m(X)\}, \qquad \hat{f} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \big\{Y_i - f(m(X_i))\big\}^2,$$

and $\mathcal{F}_{\text{iso}}$ denotes the class of nondecreasing real-valued functions. The fitted values $m^\star_{n,\text{iso}}(X_i) = \hat{f}(m(X_i))$ are blockwise empirical means determined by the pooled-adjacent-violators algorithm. This blockwise structure implies full empirical calibration, even though the calibration step is restricted to isotonic functions.

Proposition 3 (Isotonic calibration is fully calibrated)**.** *For every function $h : \mathbb{R} \to \mathbb{R}$,*

$$\mathbb{P}_n^L\big[h(m^\star_{n,\text{iso}}(X))\{Y - m^\star_{n,\text{iso}}(X)\}\big] = 0. \tag{13}$$

Moreover, it holds that

$$\mathbb{P}_n^L\big[(Y - m^\star_{n,\text{iso}}(X))^2\big] \le \mathbb{P}_n^L\big[(Y - m(X))^2\big].$$

Consequently, the AIPW representation of Theorem 1 holds with $\mathcal{F}$ equal to the class of all real-valued functions. In the context of treatment effect calibration, Proposition 3 was derived by van der Laan et al. (2023b).

Remark 3 (Implementation)**.** Isotonic regression is widely available in standard software. A standard algorithm is the pooled-adjacent-violators algorithm (PAVA) (Barlow and Brunk, 1972). Although it is generally fast, it may become slow at ultra-large sample sizes. A scalable alternative is to use regression tree software with monotonicity constraints, such as XGBoost or LightGBM (see Appendix K). In one dimension, isotonic regression is equivalent to fitting a regression tree of infinite depth under a monotonicity constraint. An advantage of tree-based software is that the fit can be regularized by constraining or tuning the number of leaves, the maximum tree depth, or the minimum number of observations per leaf.

3.3 Linear calibration via linear regression

For small labeled samples, one may prefer smoother parametric calibration rules to histogram binning or isotonic regression. Here we focus on linear calibration, which enforces that the prediction score is linearly uncorrelated with its residuals and is implemented by regressing the outcome on the prediction score (Mincer and Zarnowitz, 1969). Relative to isotonic regression, this imposes a more restrictive parametric form, akin in spirit to Platt scaling and temperature scaling (Platt et al., 1999; Guo et al., 2017), but it may be more stable when the labeled sample is small. Let

$$\big(\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a\, m(X_i) - b\}^2, \qquad m^{\star}_{n,\text{lin}}(X) := \hat{a}_{\text{lin}}\, m(X) + \hat{b}_{\text{lin}}. \tag{14}$$

Define the associated linearly calibrated plug-in estimator[4] by

$$\widehat{\psi}_{\text{lin}} := \frac{1}{n+N}\left\{\sum_{i=1}^{n} m^{\star}_{n,\text{lin}}(X_i) + \sum_{j=1}^{N} m^{\star}_{n,\text{lin}}(\widetilde{X}_j)\right\}. \tag{15}$$

More generally, one can include additional covariates in the linear regression adjustment to further improve precision, provided there is sufficient labeled data (Algorithm 3). Because the affine class includes the constant function, the least-squares fit automatically enforces empirical mean calibration. As a result, the linearly calibrated plug-in estimator admits an exact AIPW representation.

Proposition 4 (Linear calibration as AIPW)**.** *The normal equations for equation (14) imply*

$$\mathbb{P}^L_n\{Y - m^{\star}_{n,\text{lin}}(X)\} = 0 \qquad \textit{and} \qquad \mathbb{P}^L_n\big[m^{\star}_{n,\text{lin}}(X)\{Y - m^{\star}_{n,\text{lin}}(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m^{\star}_{n,\text{lin}}(X)$*,*

$$\widehat{\psi}_{\text{lin}} = \rho_n \mathbb{P}^L_n\{m^{\star}_{n,\text{lin}}(X)\} + (1-\rho_n)\mathbb{P}^U_N\{m^{\star}_{n,\text{lin}}(\widetilde{X})\} + \mathbb{P}^L_n\big[\hat{w}(X)\{Y - m^{\star}_{n,\text{lin}}(X)\}\big]. \tag{16}$$

The following proposition shows that, in the mean-estimation problem, PPI++ and linear calibration are first-order equivalent, although they are not generally identical in finite samples. In particular, linear calibration asymptotically attains the same efficiency-maximizing choice over the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$ as PPI++. Let $\widehat{\psi}_{++}(\widehat{\lambda}_{++})$ denote the PPI++ estimator of Angelopoulos et al. (2023b), that is, the AIPW estimator based on the rescaled score $\widehat{\lambda}_{++} m(X)$, where $\widehat{\lambda}_{++}$ is obtained by empirical efficiency maximization over $\{\lambda m(X) : \lambda \in \mathbb{R}\}$ (Rubin and van der Laan, 2008). The key point is that $\widehat{\lambda}_{++}$ estimates λ_0 in (4), which is precisely the population slope in the linear regression of Y on $m(X)$.

Proposition 5 (First-order equivalence of PPI++ and linear calibration)**.** *Assume* $\text{Var}\{m(X)\} > 0$*,* $\mathbb{E}[Y^2] < \infty$*, and* $\mathbb{E}[m(X)^2] < \infty$*. Then*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\text{lin}}\big)\left\{\mathbb{P}^U_N m(\widetilde{X}) - \mathbb{P}^L_n m(X)\right\}.$$

Moreover, $\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = O_p(n^{-1/2})$ *and, therefore,*

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = O_p(n^{-1/2})\, O_p\big((n+N)^{-1/2}\big) = o_p\big((n+N)^{-1/2}\big).$$

Remark 4 (Relation to prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through covariate adjustment in small-sample settings (Hansen, 2008; Rosenblum and van der Laan, 2009; Lin, 2013; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). Proposition 5 shows that PPI++ admits the same first-order interpretation.

[4] We already introduced this estimator in Remark 2. It can equivalently be implemented by balancing on $(1, m(X))$.

Algorithm 2 Linearly calibrated plug-in estimator

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{b}) \in \arg\min_{a,b} \sum_{i=1}^n \{Y_i - (aT_i + b)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a}T_i + \hat{b}$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a}\widetilde{T}_j + \hat{b}$
4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$
6: Set $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

Algorithm 3 Linearly calibrated plug-in estimator with covariate adjustment

Require: $\mathcal{D}_L = \{(X_i, Y_i)\}_{i=1}^n$, $\mathcal{D}_U = \{\widetilde{X}_j\}_{j=1}^N$, score m, level $1-\alpha$

1: Compute $T_i = m(X_i)$ and $\widetilde{T}_j = m(\widetilde{X}_j)$
2: Fit $(\hat{a}, \hat{\beta}, \hat{b}) \in \arg\min_{a,\beta,b} \sum_{i=1}^n \{Y_i - (a + X_i^\top \beta + bT_i)\}^2$
3: Set $m^\star_{n,\mathrm{lin}}(X_i) = \hat{a} + X_i^\top \hat{\beta} + \hat{b}T_i$ and $m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) = \hat{a} + \widetilde{X}_j^\top \hat{\beta} + \hat{b}\widetilde{T}_j$
4: Form $\widehat{\psi}_{\mathrm{lin}} \leftarrow (n+N)^{-1}\{\sum_{i=1}^n m^\star_{n,\mathrm{lin}}(X_i) + \sum_{j=1}^N m^\star_{n,\mathrm{lin}}(\widetilde{X}_j)\}$
5: Set $\widehat{D}_i^L = m^\star_{n,\mathrm{lin}}(X_i) - \widehat{\psi}_{\mathrm{lin}} + \rho_n^{-1}\{Y_i - m^\star_{n,\mathrm{lin}}(X_i)\}$ and $\widehat{D}_j^U = m^\star_{n,\mathrm{lin}}(\widetilde{X}_j) - \widehat{\psi}_{\mathrm{lin}}$
6: Set $\widehat{\mathrm{SE}}^2_{\mathrm{lin}} = (n+N)^{-2}\{\sum_{i=1}^n (\widehat{D}_i^L)^2 + \sum_{j=1}^N (\widehat{D}_j^U)^2\}$
7: Return $\widehat{\psi}_{\mathrm{lin}} \pm z_{1-\alpha/2}\widehat{\mathrm{SE}}_{\mathrm{lin}}$

Remark 5 (Platt scaling adjustment)**.** Linear calibration can also be achieved using losses other than squared error, such as logistic loss, Poisson loss, or, more generally, losses from canonical-link generalized linear models (Rosenblum and van der Laan, 2009). In particular, when $Y \in \{0,1\}$ and $m(X) \in (0,1)$ is an initial predictor, one may apply Platt scaling (Cox, 1958; Platt et al., 1999) by fitting a logistic regression of Y on the logit of $m(X)$:

$$\big(\hat{a}_{\mathrm{platt}}, \hat{b}_{\mathrm{platt}}\big) \in \arg\min_{a,b\in\mathbb{R}} \sum_{i=1}^n \Big[-Y_i \log \sigma\big(a \operatorname{logit}(m(X_i)) + b\big) - (1-Y_i)\log\Big(1 - \sigma\big(a \operatorname{logit}(m(X_i)) + b\big)\Big)\Big], \quad (17)$$

with calibrated predictor

$$m^\star_{n,\mathrm{platt}}(X) := \sigma\big(\hat{a}_{\mathrm{platt}} \operatorname{logit}(m(X)) + \hat{b}_{\mathrm{platt}}\big),$$

where $\sigma(t) := (1+e^{-t})^{-1}$. Platt scaling may be preferred over linear regression calibration when predicted probabilities exhibit over- or under-confidence, in which case the calibration curve is often approximately sigmoidal. More generally, outcomes can be rescaled to $[0,1]$ before applying this procedure.

3.4 Cross-Fitting and Cross-Calibration

In the main exposition, we assume for simplicity that the initial black-box predictor is fixed. In practice, however, it may be trained on the labeled sample itself, in which case the same labeled observations are reused both to fit the predictor and to calibrate its scores. A cross-fitted version avoids this reuse by replacing the single prediction function with out-of-fold predictions and then fitting the calibrator on the resulting labeled out-of-fold scores.

1. Partition the labeled sample indices into folds $I_1, \ldots, I_K$.
2. For each fold k, train the black-box regression $m^{(-k)}$ on the labeled observations outside I_k, compute out-of-fold predictions $m^{(-k)}(X_i)$ for $i \in I_k$, and compute predictions $m^{(-k)}(\widetilde{X}_j)$ on the unlabeled sample.

3. Pool the labeled out-of-fold predictions $\{m^{(-k(i))}(X_i)\}_{i=1}^n$ and fit a calibrator f_n to the labeled pairs $(m^{(-k(i))}(X_i), Y_i)$.

4. Form calibrated predictions
$$m_n^\star(X_i) = f_n(m^{(-k(i))}(X_i))$$
and
$$m_n^\star(\widetilde{X}_j) = \frac{1}{K}\sum_{k=1}^{K} f_n(m^{(-k)}(\widetilde{X}_j)),$$
and then compute the plug-in estimator equation (9).

This cross-fitted modification is the natural DML-style implementation when one wishes to reduce overfitting from estimating the initial regression on the same labeled data later used for calibration and inference (Zheng and Van Der Laan, 2010; Chernozhukov et al., 2018a). This procedure has been used for debiased inference in van der Laan et al. (2024b,c); Rabenseifner et al. (2025), and a related cross-calibration variant appears in van der Laan et al. (2023b).

4 Theory for isotonic calibration

We study the isotonic-calibrated plug-in estimator $\widehat{\psi}_{\text{iso}}$. We first establish its asymptotic linearity, asymptotic normality, and validity of Wald inference. We then study its efficiency, showing that $\widehat{\psi}_{\text{iso}}$ is efficient in a reduced data model and giving conditions under which it is also efficient in the full model, that is, attains the minimum asymptotic variance among regular estimators (Bickel et al., 1993; Van der Vaart, 2000). To keep the exposition light, we focus on the non-cross-fitted estimator and treat the initial black-box score m as fixed.

4.1 Asymptotic linearity and inference

We now state the main asymptotic expansion for $\widehat{\psi}_{\text{iso}}$. Throughout this subsection, let f_0 denote a population isotonic calibration map for the fixed score m, defined by

$$f_0 \in \arg\min_{f\in\mathcal{F}_{\text{iso}}} \mathbb{E}_{P_0}\big[\{Y - f(m(X))\}^2\big], \qquad m_0 = f_0 \circ m. \tag{18}$$

Then m_0 is the population L^2-optimal monotone transformation of $m(X)$.

Assumption 2 (Regularity conditions for isotonic calibration)**.** Assume:

(i) **Boundedness:** there exists $C_0 < \infty$ such that $|\mathbb{E}[Y \mid m(X)]| \le C_0$ almost surely and $\sup_x |m_{n,\text{iso}}^\star(x)| = O_p(1)$.

(ii) **Controlled labeled errors:** $Y - \mathbb{E}[Y \mid m(X)]$ is sub-Gaussian or subexponential under P_0.

The target of the asymptotic analysis is the population isotonic limit m_0, which need not equal the true regression μ_0. Assumption 2 ensures that the classical isotonic rate $\|m_{n,\text{iso}}^\star - m_0\|_{2,P_{0,X}}^2 = O_p(n^{-2/3})$ is achieved; see, for example, Chatterjee et al. (2015); Yang and Barber (2018).

Theorem 2 (Asymptotic linearity under the two-sample model)**.** *Suppose Assumptions 1 and 2 hold and the calibration score equations contain the constant score $h \equiv 1$. Then*

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 = \rho_n(\mathbb{P}_n^L - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})(m_0 - \psi_0) + R_{n,N},\end{aligned} \tag{19}$$

where $R_{n,N} = o_p(M^{-1/2})$ satisfies

$$R_{n,N} = O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big)$$

Thus, under the $\sqrt{M}$ scaling, the isotonic calibrated plug-in estimator is asymptotically linear with influence pair $(D^L_{m_0}, D^U_{m_0})$ given by

$$D^L_{m_0}(X,Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \qquad D^U_{m_0}(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0.$$

When $N > n$, the finite linear approximation error satisfies $R_{n,N} = O_p(n^{-2/3})$, where this rate arises from the nonparametric complexity of the isotonic calibrator f_n; under linear calibration, the corresponding remainder is typically $O_p(n^{-1})$.

Corollary 1 (Asymptotic normality and Wald inference)**.** *Under the conditions of theorem 2,*

$$\sqrt{M}\,(\widehat{\psi}_{\text{iso}} - \psi_0) \xrightarrow{d} N(0, \sigma_0^2),$$

where

$$\sigma_0^2 := \rho_0 \operatorname{Var}\left\{m_0(X) - \psi_0 + \rho_0^{-1}(Y - m_0(X))\right\} + (1-\rho_0)\operatorname{Var}\left\{m_0(\widetilde{X}) - \psi_0\right\}.$$

If

$$\widehat{D}^L_i := m^\star_{n,\text{iso}}(X_i) - \widehat{\psi}_{\text{iso}} + \rho_n^{-1}\{Y_i - m^\star_{n,\text{iso}}(X_i)\}, \qquad \widehat{D}^U_j := m^\star_{n,\text{iso}}(\widetilde{X}_j) - \widehat{\psi}_{\text{iso}},$$

and

$$\widehat{\sigma}^2 := \rho_n \mathbb{P}^L_n\big[(\widehat{D}^L)^2\big] + (1-\rho_n)\mathbb{P}^U_N\big[(\widehat{D}^U)^2\big],$$

then $\widehat{\sigma}^2 \to_p \sigma_0^2$, and standard Wald confidence intervals follow.

The above theorem is largely a special case of the calibrated DML analysis in van der Laan et al. (2024c). The proof combines the AIPW representation in Theorem 1, the finite uniform entropy integral of the isotonic calibrator class to control empirical-process remainders, and standard DML/TMLE arguments (Bickel et al., 1993; van der Laan and Robins, 2003; Bang and Robins, 2005; van der Laan and Rose, 2011); see, for example, Kennedy (2024) for a review.

4.2 Constructing confidence intervals via the bootstrap

Confidence intervals may be constructed using the standard Wald approximation from Corollary 1, namely

$$\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\,\frac{\hat{\sigma}}{\sqrt{n+N}}.$$

As an alternative, one may use the nonparametric bootstrap and refit the isotonic calibration step within each bootstrap replicate, as proposed and theoretically studied by van der Laan et al. (2024c); see also Tang and Westling (2024). This avoids direct estimation of the influence-function variance and automatically propagates uncertainty from estimation of the calibration map. `Python` code for both approaches is provided in Appendix K.

For each bootstrap replicate $b = 1, \ldots, B$, we sample with replacement from the labeled observations $\{(X_i, Y_i)\}_{i=1}^n$ and, independently, from the unlabeled observations $\{\widetilde{X}_j\}_{j=1}^N$, obtaining bootstrap samples

$$\{(X_i^{*(b)}, Y_i^{*(b)})\}_{i=1}^n, \qquad \{\widetilde{X}_j^{*(b)}\}_{j=1}^N.$$

We then refit the isotonic calibrator on the bootstrap labeled sample:

$$\hat{f}^{*(b)} \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n \big\{Y_i^{*(b)} - f(m(X_i^{*(b)}))\big\}^2,$$

and define

$$m_{\text{iso}}^{*(b)}(X) := \hat{f}^{*(b)}\{m(X)\}.$$

The corresponding bootstrap replicate of the estimator is

$$\widehat{\psi}_{\text{iso}}^{*(b)} := \frac{1}{n+N}\left\{\sum_{i=1}^{n} m_{\text{iso}}^{*(b)}(X_i^{*(b)}) + \sum_{j=1}^{N} m_{\text{iso}}^{*(b)}(\widetilde{X}_j^{*(b)})\right\}.$$

After computing B bootstrap replicates, one may form a percentile interval using the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of

$$\{\widehat{\psi}_{\text{iso}}^{*(b)}\}_{b=1}^{B}.$$

Alternatively, one may use the bootstrap standard error

$$\widehat{\sigma}_{\text{boot}}^2 := \frac{1}{B-1}\sum_{b=1}^{B}\left(\widehat{\psi}_{\text{iso}}^{*(b)} - \bar{\psi}_{\text{iso}}^{*}\right)^2, \qquad \bar{\psi}_{\text{iso}}^{*} := \frac{1}{B}\sum_{b=1}^{B}\widehat{\psi}_{\text{iso}}^{*(b)},$$

and report the normal-approximation interval

$$\widehat{\psi}_{\text{iso}} \pm z_{1-\alpha/2}\,\widehat{\sigma}_{\text{boot}}.$$

The key implementation detail is that the isotonic regression step must be refit within each bootstrap replicate rather than held fixed. Otherwise, the resulting interval does not reflect uncertainty from estimation of the calibration map.

4.3 Efficiency considerations

In this section, we clarify the efficiency properties of the isotonic-calibrated plug-in estimator $\widehat{\psi}_{\text{iso}}$. The asymptotic linearity result of Theorem 2 yields the influence pair

$$D_{m_0}^{L}(X,Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \qquad D_{m_0}^{U}(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0.$$

The key point is that $\widehat{\psi}_{\text{iso}}$ uses the covariates only through the one-dimensional score $m_0(X)$. It is therefore natural to assess its efficiency relative to a model in which $m_0(X)$, rather than the full covariate X, is observed. In that reduced two-sample model, calibration implies $\mathbb{E}[m_0(X)] = \mathbb{E}[Y] = \psi_0$, and the pair above is the efficient influence pair for ψ_0. We then compare this reduced-model benchmark with the full-model benchmark and identify conditions under which they coincide. Our semiparametric efficiency results for the two-sample setting are straightforward applications of more general data-fusion results in Li and Luedtke (2023); Graham et al. (2024).

Proposition 6 (Efficient influence pair in the reduced model indexed by m_0)**.** *Consider the reduced two-sample experiment*

$$\mathcal{E}_{m_0} := \left(\{(m_0(X_i), Y_i)\}_{i=1}^{n},\ \{m_0(\widetilde{X}_j)\}_{j=1}^{N}\right),$$

obtained by replacing the full covariate X with the scalar summary $m_0(X)$. On the submodel satisfying

$$\mathbb{E}_P[Y] = \mathbb{E}_P[m_0(X)],$$

the efficient influence pair for $\psi(P) = \mathbb{E}_P[Y]$ is

$$D_{m_0}^{L}(X,Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \tag{20}$$

$$D_{m_0}^{U}(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0. \tag{21}$$

Corollary 2 (Efficiency regimes)**.** *Under the conditions of theorem 2, the following hold.*

(i) The influence pair in theorem 2 coincides with the efficient influence pair in proposition 6. Hence $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the coarsened experiment $\mathcal{E}_{m_0}$.

(ii) If $m_0 = \mu_0$ almost surely, then the influence pair in theorem 2 coincides with the efficient influence pair in proposition 2. Hence $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the full two-sample experiment.

(iii) If $\mathbb{E}[Y \mid m(X)]$ is monotone in $m(X)$, then $m_0(X) = \mathbb{E}[Y \mid m(X)]$ almost surely. Hence $\widehat{\psi}_{\text{iso}}$ is efficient for ψ_0 in the coarsened experiment $\mathcal{E}_m$.

Remark 6 (Practical takeaway)**.** From a practical perspective, isotonic calibration can improve efficiency whenever the original prediction score is informative but miscalibrated. The estimator remains a simple plug-in estimator, but it behaves as if one had replaced the original score by its best monotone calibration.

Remark 7 (When full efficiency is attained)**.** Full efficiency requires that the calibrated target recover all relevant regression information. This holds if $m_0 = \mu_0$ almost surely, and more generally fails when the one-dimensional score $m(X)$ omits predictive information in X that cannot be recovered through monotone calibration.

5 Experiments

5.1 Simulation study

We begin with a controlled binary-outcome simulation designed to isolate the effect of score miscalibration. Let $S \sim N(0, 1)$, define

$$\mu_0(S) = \{1 + \exp(-5S)\}^{-1},$$

and draw $Y \mid S \sim \text{Bernoulli}\{\mu_0(S)\}$. To induce substantial miscalibration, we apply a stronger affine-biased monotone logit distortion,

$$m(X) = \text{clip}\left[-0.15 + 0.75\,\sigma\big\{0.8\,z(S) + 0.1\,z(S)^3 - 1\big\},\, 0.01,\, 0.99\right],$$

where $z(S) := \text{logit}\{\mu_0(S)\}$. This transformation preserves ranking information while introducing nonlinear distortion on the probability scale. We report the labeled-only mean (Naive), PPI, AIPW, linear calibration, and isotonic calibration. The labeled sample size ranges over $n \in \{50, 100, 200, 400, 800, 1200, 2400\}$, with $N/n = 1$ and $N/n = 16$. All summaries are based on 5000 Monte Carlo repetitions.

figure 1 reports the four diagnostics most directly connected to the theory in two representative unlabeled-sample regimes, $N/n = 1$ and $N/n = 16$: absolute bias, empirical Monte Carlo standard deviation, coverage, and relative efficiency versus PPI. Calibration improves efficiency by reducing the residual variance of the score-based estimator, most clearly in the variability and relative-efficiency panels. In the balanced regime $N/n = 1$, AIPW is slightly more efficient than PPI in small samples and comparable in larger samples. In the large-unlabeled regime $N/n = 16$, the two become much closer, and AIPW can be slightly less efficient under substantial misspecification of m. Linear calibration captures much of the gain already in the balanced regime, while isotonic calibration can improve further once the labeled sample is large enough to support a stable nonlinear correction, especially when unlabeled data are abundant. Coverage remains broadly reasonable throughout, although the more aggressive nonlinear calibration methods mildly undercover at smaller sample sizes.

5.2 Empirical Illustration

We evaluate the proposed calibration-based estimators on the mean-estimation benchmarks used in the original experiments of Angelopoulos et al. (2023a,b), as provided in the `ppi_py` package. Our goal is not to revisit the original benchmark comparison, but rather to understand when one-dimensional calibration of a prediction score improves semisupervised mean inference. We use the same `ppi_py` datasets and retain the labeled-sample-size grids from the original examples. For each labeled sample size n, we repeatedly split

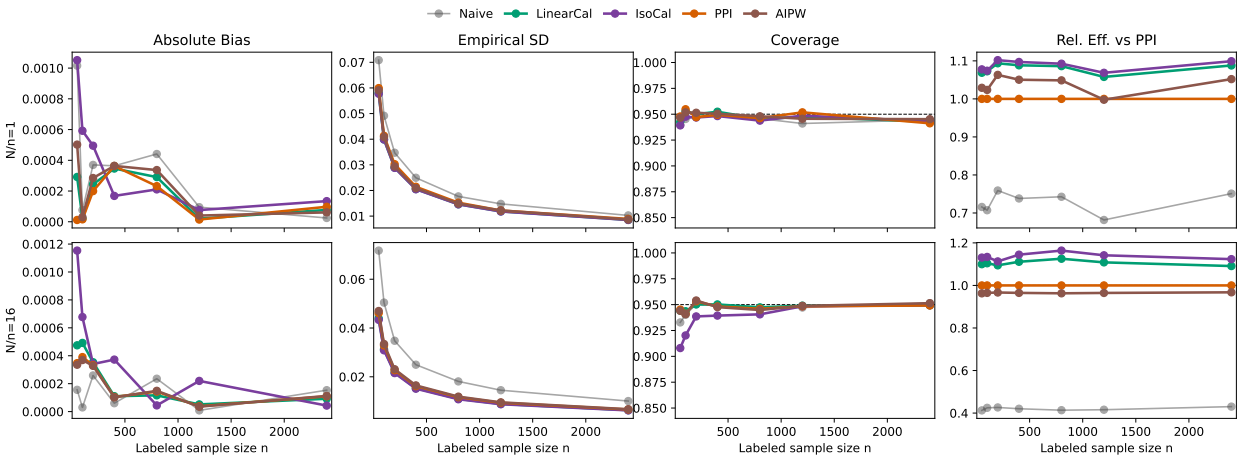

Figure 1: Synthetic simulation study with a poorly calibrated score in two unlabeled-sample regimes. The top row shows the balanced case $N/n = 1$, and the bottom row shows the large-unlabeled case $N/n = 16$. The four columns report absolute bias, empirical Monte Carlo standard deviation, coverage, and relative efficiency versus PPI, computed from empirical Monte Carlo variance. AIPW and PPI are most clearly separated in the balanced regime, while they become much closer once unlabeled data are abundant.

the fully labeled benchmark into a labeled subset of size n and an unlabeled subset of size N, apply all estimators to the same split, and treat the full-sample mean as the ground truth.[5] Results are averaged over 5000 random splits for each n.

Benchmarks and baselines. We study the `forest`, `galaxies`, and `census_income` semisupervised mean-estimation examples from the official benchmark suite. The retained baselines are the original labeled-only interval, the imputation benchmark on the binary-outcome datasets, the semisupervised linear-regression baseline from `ppi_py` that uses the full covariate vector, PPI, and the classical AIPW estimator based on the same prediction score. The imputation benchmark averages model predictions over the pooled sample without a labeled residual correction, whereas the semisupervised linear baseline fits a linear regression on the full covariates and combines unlabeled prediction averaging with a labeled residual correction. AIPW is the corresponding pooled plug-in plus augmentation estimator based on the original score, whereas PPI is a less efficient variant that discards labeled covariate information in the plug-in term.

We additionally consider three calibration-based comparators that fall directly within our framework: linear calibration (`LinearCal`), Platt scaling (`Platt`) on the binary-outcome datasets, and isotonic calibration with a fixed minimum of 10 observations per bin (`IsoCal`). For the binary-outcome datasets, we also report the Venn–Abers shrinkage variant described in Section 3.2. Platt scaling and Venn–Abers are omitted on `census_income` because the outcome is not binary. We do not include PPI++ as a separate baseline, since in this one-dimensional score setting it follows the same first-order correction path as linear calibration and classical prognostic-score adjustment. To keep the displayed figures and appendix tables readable, we omit the semisupervised linear baseline from those summaries and focus there on labeled-only, imputation, PPI, AIPW, and the calibration-based estimators.

Metrics and main findings. For each estimator, we report Monte Carlo bias, empirical variance, MSE, Wald-interval coverage, and relative efficiency versus PPI, where relative efficiency is computed from empirical Monte Carlo variances. Estimated standard errors are used only to construct Wald intervals; all reported variability summaries are empirical Monte Carlo quantities.

[5] Because repeated sample splits of a fixed finite benchmark dataset are not i.i.d. draws from a superpopulation, asymptotic Wald confidence intervals can exhibit either undercoverage or overcoverage. This is not specific to our calibration extensions, but is a feature of evaluating Wald-type intervals under finite-population resampling.

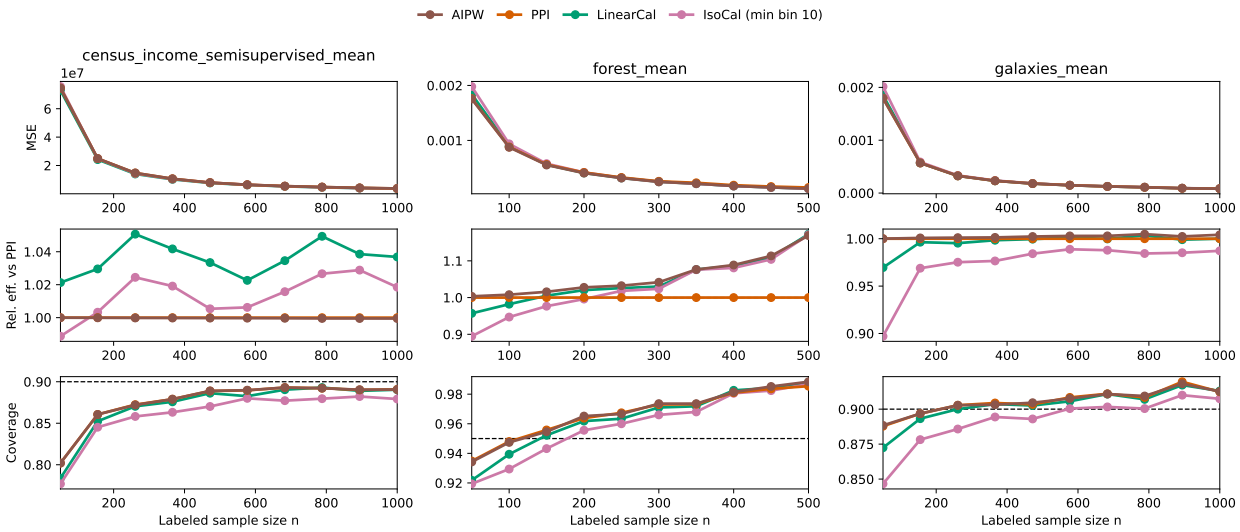

Figure 2: Main-text benchmark summary showing MSE, relative efficiency versus PPI, and coverage across the reproduced PPI benchmarks, restricted to the most informative score-based comparators: AIPW, PPI, LinearCal, and IsoCal (minimum bin size 10). Relative efficiency is computed from empirical Monte Carlo variance, so values above one indicate improvement over PPI. The dashed horizontal line in the coverage panels marks the nominal target coverage, which is 0.95 for `forest` and 0.90 for `galaxies` and `census_income`.

The results suggest a simple practical picture. AIPW is the natural estimator based on the original score, and it typically matches or outperforms PPI. In `forest`, where the original score is already strong, PPI becomes less efficient as the labeled sample grows, whereas in `galaxies` the two methods remain nearly tied. When the unlabeled sample is much larger than the labeled sample, however, AIPW and PPI become nearly indistinguishable because the labeled contribution to the pooled plug-in term is negligible; this is what we observe in `census_income`, where the unlabeled sample is enormous.

Post-hoc calibration is most useful when the score is meaningfully miscalibrated and the labeled sample is large enough to estimate the second-stage calibrator stably. In that regime, linear calibration often captures most of the available gain under mild misspecification, as in `census_income`, whereas more flexible monotone calibration is mainly competitive once the labeled sample is moderately large, with `galaxies` providing the clearest example. Overall, these results leave little practical motivation for PPI: when m is accurate, AIPW is more efficient, and when m is inaccurate, even simple calibration methods such as LinearCal often remove much of the misspecification bias.

6 Conclusion and closing remarks

We studied semisupervised mean estimation with a black-box prediction score and proposed a simple calibrated plug-in procedure: fit the score, calibrate it on the labeled sample, and average the calibrated predictions over the pooled covariate sample. Our main observation is that this seemingly simple plug-in rule also admits an exact AIPW representation, placing it directly within the standard semiparametric framework for debiased estimation. In particular, isotonic calibration yields a lightweight, tuning-free estimator with asymptotic linearity, asymptotic normality, and valid Wald inference under weak conditions. Confidence intervals may be constructed from the influence function, or equivalently from the residual variance; alternatively, the bootstrap may better capture the finite-sample variability of the isotonic regression update.

From an efficiency perspective, the isotonic-calibrated estimator is efficient for the information retained by the calibrated score $m_0(X)$. It is nonparametrically efficient for the full covariate information in X when the calibrated limit m_0 coincides with the true regression function μ_0. Isotonic calibration improves efficiency

Dataset	Smaller labeled-sample regime	Larger labeled-sample regime
`forest`	AIPW weakly dominates PPI; the added flexibility of LinearCal and IsoCal leads to higher finite-sample variance relative to AIPW.	AIPW is substantially more efficient than PPI, while additional calibration yields little beyond modest MSE gains.
`galaxies`	AIPW and PPI are very close; LinearCal is the closest calibration competitor, and IsoCal also remains competitive.	The leading score-based methods become very similar, so simple corrections are usually sufficient.
`census_income`	PPI and AIPW are effectively indistinguishable because the unlabeled sample is very large; flexible calibration can be too aggressive when the labeled sample is very small.	LinearCal performs best on average, with IsoCal also competitive once the labeled sample is large enough.

Table 1: Compact summary of the reproduced PPI benchmarks across regimes. The main empirical distinction is between small labeled-sample regimes, where simple or no calibration is safer, and larger labeled-sample regimes, where calibration can help when the score is miscalibrated.

over the simple AIPW estimator based on the original score $m(X)$, and typically also improves on linear calibration. Empirically, the practical message is simple: when the original score is already well calibrated, calibration changes little, whereas under meaningful miscalibration, even simple affine or monotone calibration can improve semisupervised efficiency while preserving a transparent plug-in form.

Positioning within existing semiparametric methods. More broadly, our results clarify that recent prediction-powered inference methods belong squarely within the standard semiparametric, debiased machine learning, and flexible covariate-adjustment toolkit for randomized trials and missing-at-random settings. In particular, we show that PPI++ is AIPW with empirical efficiency maximization (Rubin and van der Laan, 2008), and is first-order equivalent to linear calibration and classical prognostic-score regression adjustment (Hansen, 2008; Rosenblum and van der Laan, 2009; Schuler et al., 2022; Højbjerre-Frandsen et al., 2025). Thus, at least for semisupervised mean estimation, prediction-powered inference is best viewed not as a new inferential paradigm, but as a concrete instance of familiar semiparametric ideas. It is therefore closely related to randomized-trial approaches that learn prognostic scores from larger historical or auxiliary data sets and then use them for regression adjustment in smaller trials (Hansen, 2008; Højbjerre-Frandsen et al., 2025). This perspective suggests that semisupervised inference may benefit from closer connections to the broader missing-data, causal-inference, and debiased machine learning literatures, especially through translational work that adapts tools developed for randomized trials and missing-at-random settings to semisupervised problems. More generally, when missingness occurs in both outcomes and covariates and is allowed to depend on observed covariates and outcomes, as well as in extensions to right-censored time-to-event outcomes and longitudinal settings, existing semiparametric methods can be applied directly (Robins et al., 1994; van der Laan and Robins, 2003; Bang and Robins, 2005; Moore and van der Laan, 2009b; Rose and van der Laan, 2011).

Extensions. While we focus on calibration of regression functions and inference on means, the ideas here may extend to more general supervised learning problems defined through loss minimization, such as median and quantile regression (Noarov and Roth, 2023; Jung et al., 2021; Roth, 2022; Whitehouse et al., 2024). As a concrete example, Algorithm 3 extends our linearly calibrated mean estimator to linear regression by including the prediction score as a covariate, in the spirit of classical prognostic adjustment. Related validity results for such adjustments in generalized linear models are given by Rosenblum and van der Laan (2009). Similarly, for median and quantile regression, one could include prediction scores as covariates and use Wald-type inference for the corresponding target parameters. More generally, when the target parameter is defined by a moment equation, as in M-estimation, one could debias or calibrate the associated augmented moment equation; see Appendix C. A practical challenge in this setting is that augmented loss functions may be nonconvex, which can complicate optimization. In such cases, a more practical alternative may be a sieve-based, TMLE-style multiaccuracy adjustment rather than explicit augmentation, as in the Efficient Plug-in Learning framework of van der Laan et al. (2024a), which addresses this issue for nonconvex Neyman-orthogonal loss functions

(Foster and Syrgkanis, 2023). Debiased inference and influence-function theory for generic smooth functionals of an M-estimand are studied by van der Laan et al. (2025a).

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A Additional remarks

Remark 8 (Shrinkage via Venn–Abers)**.** When the labeled sample is small, isotonic calibration may be unstable. A natural alternative is Venn–Abers calibration, originally introduced for binary classification (Vovk and Petej, 2012); see van der Laan and Alaa (2024, 2025) for generalized versions beyond the binary setting. Assume $Y \in [0,1]$, or rescale otherwise. For each x, augment the labeled calibration sample with the hypothetical point $(m(x), y)$ for $y \in \{0,1\}$, fit the resulting isotonic calibrators $f_n^{(x,0)}$ and $f_n^{(x,1)}$, and form the interval-valued prediction

$$\big[f_n^{(x,0)}\{m(x)\},\ f_n^{(x,1)}\{m(x)\}\big].$$

Following Vovk and Petej (2012), one may convert this interval to a point prediction by shrinking its midpoint toward the (unadjusted) AIPW estimator:

$$m^*_{n,\mathrm{VA}}(x) := m^*_{n,\mathrm{mid}}(x) + \big\{f_n^{(x,1)}\{m(x)\} - f_n^{(x,0)}\{m(x)\}\big\}\big\{\widehat{\psi}_{AIPW} - m^*_{n,\mathrm{mid}}(x)\big\},$$

where

$$m^*_{n,\mathrm{mid}}(x) := \frac{f_n^{(x,1)}\{m(x)\} + f_n^{(x,0)}\{m(x)\}}{2}.$$

Since the shrinkage magnitude is proportional to the width of the Venn–Abers interval, this construction can be more stable than plain isotonic calibration in small samples.

Remark 9 (Relation between linear calibration, PPI, and prognostic-score adjustment)**.** Linear calibration can be viewed as prognostic-score regression adjustment, a classical approach to improving precision through flexible covariate adjustment in small-sample settings (Hansen, 2008; Rosenblum and van der Laan, 2009; Lin, 2013; Schuler et al., 2022; Balzer et al., 2024; Højbjerre-Frandsen et al., 2025). Proposition 5 shows that the PPI++ estimator admits the same first-order interpretation. Likewise, the PPI estimator $\widehat{\psi}_{\mathrm{PPI}}$ is algebraically equivalent to the unlabeled-sample plug-in estimator $\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$ based on the intercept-only adjustment

$$\hat{a}_{\mathrm{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^\star_{n,\mathrm{const}}(X) := \hat{a}_{\mathrm{const}} + m(X), \tag{22}$$

which calibrates only the mean. By contrast, the standard AIPW estimator $\widehat{\psi}_{\mathrm{AIPW}}$ is equivalent to the pooled-sample mean $\rho_n\mathbb{P}_n^L\{m^\star_{n,\mathrm{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\mathrm{const}}(X)\}$.

B Prediction-Powered Causal Inference via Calibration (i.e., Calibrated DML)

This appendix records the direct translation of the two-sample missing-outcome setup to randomized experiments. The central point is that the semisupervised problem studied in the main text and two-arm causal inference in a randomized trial are the same statistical problem once the unobserved potential outcome in one arm is interpreted as the missing outcome. We then illustrate how our calibrated plug-in estimators apply in this setting, where they arise as special cases of calibrated DML (van der Laan et al., 2024c).

We use the standard potential-outcomes notation. Each unit has two potential outcomes, $Y(1)$ under treatment and $Y(0)$ under control, and the observed outcome satisfies

$$Y = AY(1) + (1-A)Y(0),$$

where $A \in \{0,1\}$ is the treatment assignment. The mean potential outcomes are

$$\mu_1 := \mathbb{E}\{Y(1)\}, \qquad \mu_0 := \mathbb{E}\{Y(0)\},$$

and the average treatment effect is $\tau := \mu_1 - \mu_0$. In a randomized trial, identification follows from the usual causal conditions: consistency and no interference, exchangeability induced by randomization,

$$A \perp (X, Y(1), Y(0)),$$

and positivity, so that each arm has positive assignment probability.

Now partition the observed data into the two arm-specific samples

$$\mathcal{D}_{n,1} := \{(X_i, Y_i) : A_i = 1\}, \qquad \mathcal{D}_{n,0} := \{(\widetilde{X}_j, \widetilde{Y}_j) : A_j = 0\},$$

with sample sizes $n_1 := |\mathcal{D}_{n,1}|$, $n_0 := |\mathcal{D}_{n,0}|$, and $n = n_1 + n_0$. Let $\mathbb{P}_{n,a}$ denote the empirical mean over arm a, and let $\hat{\pi}_a := n_a/n$. To estimate μ_1, one treats $\mathcal{D}_{n,1}$ as the labeled sample for $Y(1)$ and uses the covariates in $\mathcal{D}_{n,0}$ as the unlabeled sample. To estimate μ_0, one reverses the roles of the two arms. Thus each mean potential outcome is a two-sample missing-outcome estimand of exactly the form analyzed in the main text.

For $a \in \{0,1\}$, let $m_a(X)$ be an arm-specific prediction score fit using only observations in arm a. The standard arm-specific AIPW estimator is

$$\hat{\mu}_a^{\mathrm{AIPW}} := \mathbb{P}_n\{m_a(X)\} + \mathbb{P}_n\left[\frac{\mathbb{1}(A=a)}{\hat{\pi}_a}\{Y - m_a(X)\}\right].$$

For $a \in \{0,1\}$, let $m^*_{n,a}(X)$ denote the calibrated version of the score $m_a(X)$. The calibrated estimator of μ_a is then

$$\hat{\mu}_a^{\text{cal}} := \mathbb{P}_n\{m^*_{n,a}(X)\} = \hat{\pi}_a\, \mathbb{P}_{n,a}\{m^*_{n,a}(X)\} + (1-\hat{\pi}_a)\, \mathbb{P}_{n,1-a}\{m^*_{n,a}(X)\}. \tag{23}$$

This is exactly the pooled plug-in estimator from the two-sample formulation, with one arm contributing outcomes and the other contributing only covariates. The same estimator also has the exact AIPW form

$$\hat{\mu}_a^{\text{cal}} = \mathbb{P}_n\{m^*_{n,a}(X)\} + \mathbb{P}_n\left[\frac{\mathbb{1}(A=a)}{\hat{\pi}_a}\{Y - m^*_{n,a}(X)\}\right]. \tag{24}$$

Because

$$\mathbb{P}_n\left[\frac{\mathbb{1}(A=a)}{\hat{\pi}_a}\{Y - m^*_{n,a}(X)\}\right] = \mathbb{P}_{n,a}\{Y - m^*_{n,a}(X)\},$$

the residual correction vanishes exactly whenever the calibrator includes the constant score and therefore enforces mean calibration within arm a. This is the randomized-trial analogue of the exact augmented representation in theorem 1.

The average treatment effect is then estimated by

$$\hat{\tau}^{\text{cal}} := \hat{\mu}_1^{\text{cal}} - \hat{\mu}_0^{\text{cal}}. \tag{25}$$

Equivalently,

$$\hat{\tau}^{\text{cal}} = \mathbb{P}_n\{\tilde{m}_1(X) - \tilde{m}_0(X)\} + \mathbb{P}_n\left[\frac{A}{\hat{\pi}_1}\{Y - \tilde{m}_1(X)\} - \frac{1-A}{\hat{\pi}_0}\{Y - \tilde{m}_0(X)\}\right].$$

Its influence function is the difference of the two arm-specific AIPW influences,

$$D_\tau(O) = \left\{\tilde{m}_1(X) - \mu_1 + \frac{A}{\pi_1}\big(Y - \tilde{m}_1(X)\big)\right\} - \left\{\tilde{m}_0(X) - \mu_0 + \frac{1-A}{\pi_0}\big(Y - \tilde{m}_0(X)\big)\right\}.$$

Accordingly, Wald inference for the ATE may be based on the empirical variance of the estimated influence values divided by n, or on the nonparametric bootstrap refitting the calibration step within each bootstrap sample.

The practical implementation is therefore immediate. One splits the trial into $\mathcal{D}_{n,1}$ and $\mathcal{D}_{n,0}$, treats one arm as labeled and the other as unlabeled when estimating a given mean potential outcome, fits and calibrates an arm-specific score, averages the calibrated predictions over the pooled covariate sample, and finally subtracts the two arm-specific estimates to obtain the ATE. In this way, the semisupervised mean estimators in the main text become direct estimators of mean potential outcomes and treatment effects in randomized experiments.

C General Moment Equations Under MCAR

We briefly review the general AIPW class for moment-equation targets under missing completely at random; see, for example, Robins et al. (1995); van der Laan and Robins (2003). Let

$$O = (X, R, RY),$$

where $R \in \{0,1\}$ indicates whether Y is observed, and suppose

$$R \perp Y \mid X, \qquad \pi_0(X) := \mathbb{P}(R=1 \mid X) > 0 \text{ a.s.}$$

Let the target $\theta_0 \in \Theta \subseteq \mathbb{R}^d$ be identified by

$$\mathbb{E}[\varphi(Y, \theta_0)] = 0,$$

where $\varphi : \mathcal{Y} \times \Theta \to \mathbb{R}^d$ is a known identifying function. Define

$$A_0 := \partial_\theta \mathbb{E}[\varphi(Y,\theta)]\big|_{\theta=\theta_0},$$

and assume A_0 is nonsingular.

Theorem 3 (AIPW class for general moment equations under MCAR)**.** *For any square-integrable function $m : \mathcal{X} \times \Theta \to \mathbb{R}^d$, define*

$$\psi(O; \theta, m, \pi_0) := m(X, \theta) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta) - m(X, \theta)\}. \tag{26}$$

Then

$$\mathbb{E}[\psi(O; \theta_0, m, \pi_0)] = 0.$$

Hence any estimator $\hat{\theta}$ solving the empirical estimating equation

$$\mathbb{P}_n \psi(O; \hat{\theta}, m, \pi_0) = 0$$

is a valid AIPW-style estimator of θ_0.

Moreover, the corresponding influence function is

$$D_m(O) = -A_0^{-1}\psi(O; \theta_0, m, \pi_0) = -A_0^{-1}\left[m(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m(X, \theta_0)\}\right]. \tag{27}$$

Thus the class of regular influence-function-based estimators is indexed by the choice of augmentation function m.

The efficient choice in the full MCAR model is

$$m_0(X, \theta_0) := \mathbb{E}[\varphi(Y, \theta_0) \mid X], \tag{28}$$

which yields the efficient influence function

$$D_{\text{eff}}(O) = -A_0^{-1}\left[m_0(X, \theta_0) + \frac{R}{\pi_0(X)}\{\varphi(Y, \theta_0) - m_0(X, \theta_0)\}\right]. \tag{29}$$

Corollary 3 (Two-sample/PPI form)**.** *Suppose the labeled and unlabeled samples arise from the semisupervised model with constant labeling probability ρ_0, and let $S = f(X)$ be a chosen score. Replacing X by S gives the reduced AIPW class*

$$\psi(O; \theta, m, \rho_0) = m(S, \theta) + \frac{R}{\rho_0}\{\varphi(Y, \theta) - m(S, \theta)\}.$$

Equivalently, the sample estimating equation can be written as

$$0 = \rho_n \mathbb{P}_n^L\{m(S, \theta)\} + (1 - \rho_n)\mathbb{P}_N^U\{m(S, \theta)\} + \mathbb{P}_n^L\{\varphi(Y, \theta) - m(S, \theta)\}.$$

The efficient choice within the reduced score model is

$$m_0(S, \theta_0) = \mathbb{E}[\varphi(Y, \theta_0) \mid S].$$

D Proof of the representation theorem

Proof of theorem 1. Because equation (8) holds for every $h \in \mathcal{F}$, any empirical balancing weight $\widehat{w}(m_n^\star(X)) \in \mathcal{F}$ satisfies

$$\mathbb{P}_n^L\left[\widehat{w}(m_n^\star(X))\{Y - m_n^\star(X)\}\right] = 0.$$

Adding this zero term to the plug-in estimator equation (9) yields equation (10).

For the second claim, since the identity map belongs to $\mathcal{F}$, taking $f(t) = t$ in equation (11) gives

$$\mathbb{P}_n^L\left[\widehat{w}_n^*(m_n^\star(X))\, m_n^\star(X)\right] = \rho_n \mathbb{P}_n^L\{m_n^\star(X)\} + (1 - \rho_n)\mathbb{P}_N^U\{m_n^\star(\widetilde{X})\} = \widehat{\psi}_{\text{cal}}.$$

Since $\widehat{w}_n^* \in \mathcal{F}$, equation (8) also gives

$$\mathbb{P}_n^L\left[\widehat{w}_n^*(m_n^\star(X))\{Y - m_n^\star(X)\}\right] = 0.$$

Therefore,

$$\mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))Y\big] = \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\, m_n^\star(X)\big] + \mathbb{P}_n^L\big[\widehat{w}_n^*(m_n^\star(X))\{Y - m_n^\star(X)\}\big] = \widehat{\psi}_{\text{cal}},$$

which is equation (12). □

Lemma 1 (Blockwise residual orthogonality)**.** *Let $B_1, \ldots, B_J$ be the pooled adjacent violator blocks of the isotonic fit, and let $m_{n,\text{iso}}^\star(X_i) = c_j$ for $i \in B_j$. Then*

$$\sum_{i \in B_j} (Y_i - c_j) = 0, \qquad j = 1, \ldots, J.$$

Consequently, for every measurable function $h : \mathbb{R} \to \mathbb{R}$ such that $h(m_{n,\text{iso}}^\star(X_i))$ is finite for all i,

$$\sum_{i=1}^n h(m_{n,\text{iso}}^\star(X_i))\{Y_i - m_{n,\text{iso}}^\star(X_i)\} = 0.$$

Proof. The fitted values from isotonic regression are constant on pooled adjacent violator blocks. The KKT conditions imply that residuals sum to zero on each block. Since $h(m_{n,\text{iso}}^\star(X_i)) = h(c_j)$ for all $i \in B_j$,

$$\sum_{i=1}^n h(m_{n,\text{iso}}^\star(X_i))\{Y_i - m_{n,\text{iso}}^\star(X_i)\} = \sum_{j=1}^J h(c_j) \sum_{i \in B_j} (Y_i - c_j) = 0.$$

□

Proof of proposition 3. The orthogonality claim equation (13) is exactly the conclusion of Lemma 1. For the risk comparison, note that the identity map belongs to $\mathcal{F}_{\text{iso}}$. Since $\hat{f}$ minimizes the empirical squared loss over $\mathcal{F}_{\text{iso}}$,

$$\mathbb{P}_n^L\big[(Y - m_{n,\text{iso}}^\star(X))^2\big] \le \mathbb{P}_n^L\big[(Y - \hat{f}_{\text{id}}(m(X)))^2\big] = \mathbb{P}_n^L\big[(Y - m(X))^2\big],$$

where $\hat{f}_{\text{id}}(t) = t$. □

Proof of proposition 4. The normal equations for equation (14) are

$$\mathbb{P}_n^L\{Y - m_{n,\text{lin}}^\star(X)\} = 0 \qquad \text{and} \qquad \mathbb{P}_n^L\big[m(X)\{Y - m_{n,\text{lin}}^\star(X)\}\big] = 0.$$

Because $m_{n,\text{lin}}^\star(X) = \hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}$, these imply

$$\mathbb{P}_n^L\big[m_{n,\text{lin}}^\star(X)\{Y - m_{n,\text{lin}}^\star(X)\}\big] = 0.$$

Hence, for any $\hat{w}(X) = a + b\, m_{n,\text{lin}}^\star(X)$,

$$\mathbb{P}_n^L\big[\hat{w}(X)\{Y - m_{n,\text{lin}}^\star(X)\}\big] = a\, \mathbb{P}_n^L\{Y - m_{n,\text{lin}}^\star(X)\} + b\, \mathbb{P}_n^L\big[m_{n,\text{lin}}^\star(X)\{Y - m_{n,\text{lin}}^\star(X)\}\big] = 0.$$

Adding this zero term to equation (15) yields equation (16). □

E Proofs of first-order equivalence between PPI++ and linear calibration

Recall

$$(\hat{a}_{\text{lin}}, \hat{b}_{\text{lin}}) \in \arg\min_{a,b \in \mathbb{R}} \sum_{i=1}^n \{Y_i - a\, m(X_i) - b\}^2,$$

and define the corresponding linear-calibration estimator by

$$\widehat{\psi}_{\text{lin}} = \rho_n \mathbb{P}_n^L\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\} + (1 - \rho_n)\mathbb{P}_N^U\{\hat{a}_{\text{lin}} m(X) + \hat{b}_{\text{lin}}\}.$$

Also let $\widehat{\psi}_{++}$ denote the PPI++ estimator obtained by minimizing the empirical influence-function variance over the scaling class $\{\lambda m(X) : \lambda \in \mathbb{R}\}$, and let $\hat{\lambda}_{++}$ be the resulting selected coefficient. Define

$$\Delta_n := \mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X).$$

Finally, define

$$a_0 := \frac{\mathrm{Cov}\{Y, m(X)\}}{\mathrm{Var}\{m(X)\}}.$$

Proof of Proposition 5. The normal equations for the linear regression fit are

$$\mathbb{P}_n^L\{Y - \hat{a}_{\mathrm{lin}} m(X) - \hat{b}_{\mathrm{lin}}\} = 0$$

and

$$\mathbb{P}_n^L\big[m(X)\{Y - \hat{a}_{\mathrm{lin}} m(X) - \hat{b}_{\mathrm{lin}}\}\big] = 0.$$

The first equation gives

$$\hat{b}_{\mathrm{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\mathrm{lin}} \mathbb{P}_n^L m(X).$$

Substituting this into the second equation yields

$$\hat{a}_{\mathrm{lin}} = \frac{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))(Y - \mathbb{P}_n^L Y)\big]}{\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big]}.$$

Hence $\hat{a}_{\mathrm{lin}}$ is the ordinary least-squares slope, so under finite second moments,

$$\hat{a}_{\mathrm{lin}} = a_0 + O_p(n^{-1/2}).$$

It remains to compare the two estimators. Using $\hat{b}_{\mathrm{lin}} = \mathbb{P}_n^L Y - \hat{a}_{\mathrm{lin}} \mathbb{P}_n^L m(X)$, we have

$$\begin{aligned}
\widehat{\psi}_{\mathrm{lin}} &= \rho_n \mathbb{P}_n^L\{\hat{a}_{\mathrm{lin}} m(X) + \hat{b}_{\mathrm{lin}}\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{a}_{\mathrm{lin}} m(\widetilde{X}) + \hat{b}_{\mathrm{lin}}\} \\
&= \hat{b}_{\mathrm{lin}} + \hat{a}_{\mathrm{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y - \hat{a}_{\mathrm{lin}} \mathbb{P}_n^L m(X) + \hat{a}_{\mathrm{lin}}\Big[\rho_n \mathbb{P}_n^L m(X) + (1-\rho_n)\mathbb{P}_N^U m(\widetilde{X})\Big] \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{a}_{\mathrm{lin}}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\}.
\end{aligned}$$

Thus, linear calibration depends on the data only through the correction coefficient $(1-\rho_n)\hat{a}_{\mathrm{lin}}$ multiplying the contrast Δ_n.

On the other hand,

$$\begin{aligned}
\widehat{\psi}_{++}(\hat{\lambda}_{++}) &= \rho_n \mathbb{P}_n^L\{\hat{\lambda}_{++} m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{\hat{\lambda}_{++} m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - \hat{\lambda}_{++} m(X)\} \\
&= \mathbb{P}_n^L Y + (1-\rho_n)\hat{\lambda}_{++}\big\{\mathbb{P}_N^U m(\widetilde{X}) - \mathbb{P}_n^L m(X)\big\}.
\end{aligned}$$

Therefore,

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\mathrm{lin}} = (1-\rho_n)\big(\hat{\lambda}_{++} - \hat{a}_{\mathrm{lin}}\big)\Delta_n.$$

In particular,

$$\hat{\lambda}_{++} = \frac{\mathbb{P}_n^L\big[(Y - \mathbb{P}_n^L Y)(m(X) - \mathbb{P}_n^L m(X))\big]}{(1-\rho_n)\mathbb{P}_n^L\big[(m(X) - \mathbb{P}_n^L m(X))^2\big] + \rho_n \mathbb{P}_N^U\big[(m(\widetilde{X}) - \mathbb{P}_N^U m(\widetilde{X}))^2\big]},$$

which estimates the population minimizer of the asymptotic variance over the scaling class,

$$\lambda_0 = \frac{\mathrm{Cov}\{Y, m(X)\}}{\mathrm{Var}\{m(X)\}} = a_0.$$

Standard M-estimation arguments then imply

$$\hat{\lambda}_{++} = \lambda_0 + O_p(n^{-1/2}).$$

Moreover, since $\mathbb{P}_n^L m(X)$ and $\mathbb{P}_N^U m(\widetilde{X})$ are sample means under the same marginal law of X and $\rho_n \to \rho_0 \in (0,1)$,

$$\Delta_n = O_p\big((n+N)^{-1/2}\big).$$

Combining this with $\hat{a}_{\text{lin}} = a_0 + O_p(n^{-1/2})$ yields

$$\hat{\lambda}_{++} - \hat{a}_{\text{lin}} = O_p(n^{-1/2}),$$

and therefore

$$\widehat{\psi}_{++}(\hat{\lambda}_{++}) - \widehat{\psi}_{\text{lin}} = (1-\rho_n)\, O_p(n^{-1/2})\, O_p\big((n+N)^{-1/2}\big) = o_p\big((n+N)^{-1/2}\big).$$

This proves the claim. □

Proposition 7 (PPI and AIPW as intercept-only calibration)**.** *Let*

$$\hat{a}_{\text{const}} \in \arg\min_{a\in\mathbb{R}} \sum_{i=1}^{n} \{Y_i - a - m(X_i)\}^2, \qquad m^\star_{n,\text{const}}(X) := \hat{a}_{\text{const}} + m(X).$$

Then

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Consequently, the PPI estimator

$$\widehat{\psi}_{\text{PPI}} := \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\text{PPI}} = \mathbb{P}_N^U\{m^\star_{n,\text{const}}(\widetilde{X})\},$$

while the standard AIPW estimator

$$\widehat{\psi}_{\text{AIPW}} := \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\}$$

satisfies

$$\widehat{\psi}_{\text{AIPW}} = \rho_n \mathbb{P}_n^L\{m^\star_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\text{const}}(\widetilde{X})\}.$$

Hence PPI is the unlabeled-only plug-in estimator based on the mean-calibrated score $m^\star_{n,\text{const}}$, whereas AIPW is the corresponding pooled plug-in estimator.

Proof. The objective in the definition of $\hat{a}_{\text{const}}$ is quadratic in a, and its first-order condition is

$$0 = \mathbb{P}_n^L\{Y - a - m(X)\}.$$

Therefore

$$\hat{a}_{\text{const}} = \mathbb{P}_n^L\{Y - m(X)\}.$$

Using $m^\star_{n,\text{const}}(\widetilde{X}) = m(\widetilde{X}) + \hat{a}_{\text{const}}$, we obtain

$$\mathbb{P}_N^U\{m^\star_{n,\text{const}}(\widetilde{X})\} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} = \mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} = \widehat{\psi}_{\text{PPI}}.$$

Likewise,

$$\begin{aligned}
\rho_n \mathbb{P}_n^L\{m^\star_{n,\text{const}}(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m^\star_{n,\text{const}}(\widetilde{X})\} &= \rho_n \mathbb{P}_n^L\{m(X) + \hat{a}_{\text{const}}\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X}) + \hat{a}_{\text{const}}\} \\
&= \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \hat{a}_{\text{const}} \\
&= \rho_n \mathbb{P}_n^L\{m(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m(\widetilde{X})\} + \mathbb{P}_n^L\{Y - m(X)\} \\
&= \widehat{\psi}_{\text{AIPW}}.
\end{aligned}$$

This proves the claim. □

F Proofs of the efficient influence pairs

Proof of proposition 2. Fix P_0, and write $P_{0,X}$ for its X-marginal and $\mu_0(x) := \mathbb{E}_{P_0}[Y \mid X = x]$. The observed two-sample experiment is

$$\mathbb{D}^{(n,N)}(P) = P^{\otimes n} \otimes P_X^{\otimes N}.$$

In the local asymptotic limit with labeled fraction $\rho_0 \in (0,1)$, the relevant Hilbert space is

$$\mathbb{H} := L_0^2(P_0) \times L_0^2(P_{0,X}),$$

equipped with inner product

$$\big\langle (u^L, u^U), (v^L, v^U) \big\rangle = \rho_0 \, \mathbb{E}_{P_0}[u^L v^L] + (1-\rho_0) \, \mathbb{E}_{P_{0,X}}[u^U v^U].$$

Let P_ε be a regular parametric submodel through P_0, and factorize

$$p_\varepsilon(x,y) = p_{X,\varepsilon}(x) \, p_{Y|X,\varepsilon}(y \mid x).$$

Write

$$a(x) := \partial_\varepsilon \log p_{X,\varepsilon}(x)\big|_{\varepsilon=0}, \qquad b(x,y) := \partial_\varepsilon \log p_{Y|X,\varepsilon}(y \mid x)\big|_{\varepsilon=0}.$$

Then

$$\mathbb{E}_{P_{0,X}}[a(X)] = 0, \qquad \mathbb{E}_{P_0}[b(X,Y) \mid X] = 0,$$

and the induced score in the two-sample experiment is

$$\big(a(X) + b(X,Y),\, a(\widetilde{X})\big).$$

Hence the tangent space is

$$\mathcal{T}_{\text{full}} = \Big\{ (a+b,\ a) : a \in L_0^2(P_{0,X}),\ b \in L_0^2(P_0),\ \mathbb{E}[b(X,Y) \mid X] = 0 \Big\}.$$

The parameter is

$$\psi(P) = \mathbb{E}_P[Y] = \mathbb{E}_{P_X}[\mu_P(X)].$$

Its pathwise derivative along P_ε is

$$\partial_\varepsilon \psi(P_\varepsilon)\big|_{\varepsilon=0} = \mathbb{E}_{P_0}[Y\{a(X) + b(X,Y)\}].$$

Using the tower property,

$$\mathbb{E}_{P_0}[Y a(X)] = \mathbb{E}_{P_{0,X}}[\mu_0(X) a(X)],$$

so

$$\dot{\psi}_{P_0}(a,b) = \mathbb{E}_{P_{0,X}}[\mu_0(X) a(X)] + \mathbb{E}_{P_0}[Y \, b(X,Y)].$$

We claim that the Riesz representer of this linear functional on $\mathcal{T}_{\text{full}}$ is

$$D_{\text{eff}}^L(X,Y) = \mu_0(X) - \psi_0 + \rho_0^{-1}\{Y - \mu_0(X)\}, \qquad D_{\text{eff}}^U(\widetilde{X}) = \mu_0(\widetilde{X}) - \psi_0.$$

Indeed, both components are mean zero, and for any $(a+b, a) \in \mathcal{T}_{\text{full}}$,

$$\begin{aligned}
&\big\langle (D_{\text{eff}}^L, D_{\text{eff}}^U), (a+b, a) \big\rangle \\
&= \rho_0 \, \mathbb{E}_{P_0}\big[\big\{\mu_0(X) - \psi_0 + \rho_0^{-1}(Y - \mu_0(X))\big\} \{a(X) + b(X,Y)\}\big] + (1-\rho_0) \, \mathbb{E}_{P_{0,X}}[(\mu_0(X) - \psi_0) a(X)].
\end{aligned}$$

Now

$$\mathbb{E}_{P_0}[(\mu_0(X) - \psi_0) b(X,Y)] = 0$$

because $\mathbb{E}[b(X,Y) \mid X] = 0$, and

$$\mathbb{E}_{P_0}[(Y - \mu_0(X)) a(X)] = 0$$

because $\mathbb{E}[Y - \mu_0(X) \mid X] = 0$. Therefore

$$\begin{aligned}\big\langle (D^L_{\text{eff}}, D^U_{\text{eff}}), (a+b, a) \big\rangle &= \mathbb{E}_{P_{0,X}}[(\mu_0(X) - \psi_0)a(X)] + \mathbb{E}_{P_0}[(Y - \mu_0(X))b(X,Y)] \\ &= \mathbb{E}_{P_{0,X}}[\mu_0(X)a(X)] + \mathbb{E}_{P_0}[Y\, b(X,Y)] \\ &= \dot{\psi}_{P_0}(a,b).\end{aligned}$$

Thus $(D^L_{\text{eff}}, D^U_{\text{eff}})$ is the canonical gradient, hence the efficient influence pair. This proves equations (5) and (6). □

Proof of proposition 6. Let

$$S := m_0(X), \qquad \eta_0(s) := \mathbb{E}[Y \mid S = s].$$

Write Q_0 for the law of (S, Y), and $Q_{0,S}$ for the marginal law of S. In the reduced two-sample experiment, the labeled sample is i.i.d. from Q_0 and the unlabeled sample is i.i.d. from $Q_{0,S}$. The parameter remains

$$\psi(Q) = \mathbb{E}_Q[Y].$$

As in the full-model case, the Hilbert space is

$$\mathbb{H}_{\text{red}} := L^2_0(Q_0) \times L^2_0(Q_{0,S}),$$

with inner product

$$\big\langle (u^L, u^U), (v^L, v^U) \big\rangle = \rho_0\, \mathbb{E}_{Q_0}[u^L v^L] + (1-\rho_0)\, \mathbb{E}_{Q_{0,S}}[u^U v^U].$$

Let Q_ε be a regular submodel through Q_0, and factorize

$$q_\varepsilon(s,y) = q_{S,\varepsilon}(s)\, q_{Y|S,\varepsilon}(y \mid s).$$

Define the marginal and conditional scores

$$a(s) := \partial_\varepsilon \log q_{S,\varepsilon}(s)\big|_{\varepsilon=0}, \qquad b(s,y) := \partial_\varepsilon \log q_{Y|S,\varepsilon}(y \mid s)\big|_{\varepsilon=0}.$$

Then

$$\mathbb{E}_{Q_{0,S}}[a(S)] = 0, \qquad \mathbb{E}_{Q_0}[b(S,Y) \mid S] = 0,$$

and the tangent space is

$$\mathcal{T}_{\text{red}} = \Big\{ (a+b,\ a) : a \in L^2_0(Q_{0,S}),\ b \in L^2_0(Q_0),\ \mathbb{E}[b(S,Y) \mid S] = 0 \Big\}.$$

The pathwise derivative of $\psi(Q) = \mathbb{E}_Q[Y]$ is

$$\partial_\varepsilon \psi(Q_\varepsilon)\big|_{\varepsilon=0} = \mathbb{E}_{Q_0}[Y\{a(S) + b(S,Y)\}] = \mathbb{E}_{Q_{0,S}}[\eta_0(S)a(S)] + \mathbb{E}_{Q_0}[Y\, b(S,Y)].$$

Therefore the canonical gradient in the reduced model is

$$D^L_{\text{red}}(S,Y) = \eta_0(S) - \psi_0 + \rho_0^{-1}\{Y - \eta_0(S)\}, \qquad D^U_{\text{red}}(\widetilde{S}) = \eta_0(\widetilde{S}) - \psi_0.$$

The verification is identical to the full-model case: for any $(a+b, a) \in \mathcal{T}_{\text{red}}$,

$$\begin{aligned}&\big\langle (D^L_{\text{red}}, D^U_{\text{red}}), (a+b, a) \big\rangle \\ &= \rho_0\, \mathbb{E}_{Q_0}\big[\big\{\eta_0(S) - \psi_0 + \rho_0^{-1}(Y - \eta_0(S))\big\}\, \{a(S) + b(S,Y)\}\big] + (1-\rho_0)\, \mathbb{E}_{Q_{0,S}}[(\eta_0(S) - \psi_0)a(S)] \\ &= \mathbb{E}_{Q_{0,S}}[\eta_0(S)a(S)] + \mathbb{E}_{Q_0}[Y\, b(S,Y)],\end{aligned}$$

since

$$\mathbb{E}[(\eta_0(S) - \psi_0)b(S,Y)] = 0, \qquad \mathbb{E}[(Y - \eta_0(S))a(S)] = 0.$$

This equals the pathwise derivative, so the displayed pair is the efficient influence pair in the reduced model.

Finally, at the calibrated truth we have

$$\eta_0(S) = \mathbb{E}[Y \mid S] = S = m_0(X) \qquad \text{a.s.}$$

Substituting $S = m_0(X)$ gives

$$D^L_{m_0}(X, Y) = m_0(X) - \psi_0 + \rho_0^{-1}\{Y - m_0(X)\}, \qquad D^U_{m_0}(\widetilde{X}) = m_0(\widetilde{X}) - \psi_0,$$

which is exactly equations (20) and (21). □

Lemma 2 (Empirical calibration centers the limit)**.** *Suppose Assumptions 1 and 2 hold and the calibration score equations contain the constant score $h \equiv 1$. Then*

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

Proof. Let $T := m(X)$, and write $m_0 = f_0(T)$, where

$$f_0 \in \arg\min_{f \in \mathcal{F}_{\text{iso}}} \mathbb{E}_{P_0}\big[(Y - f(T))^2\big].$$

Because $\mathcal{F}_{\text{iso}}$ is closed under addition of constants, for every $c \in \mathbb{R}$ the function $f_0 + c$ also belongs to $\mathcal{F}_{\text{iso}}$. Hence the function

$$g(c) := \mathbb{E}_{P_0}\big[(Y - f_0(T) - c)^2\big]$$

is minimized at $c = 0$. Since g is differentiable,

$$0 = g'(0) = -2\,\mathbb{E}_{P_0}[Y - f_0(T)].$$

Therefore

$$\mathbb{E}_{P_0}[Y] = \mathbb{E}_{P_0}[f_0(T)].$$

Recalling that $m_0(X) = f_0(m(X)) = f_0(T)$ and $\psi_0 = \mathbb{E}_{P_0}[Y]$, we conclude

$$\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0.$$

□

Proof convention. In the following proofs, fix $\delta > 0$. By $\sup_x |m^\star_{n,\text{iso}}(x)| = O_p(1)$, there exist a deterministic constant $C = C_\delta \geq C_0$ and $n_0 < \infty$ such that the event

$$A_{n,C} := \Big\{\sup_x |m^\star_{n,\text{iso}}(x)| \leq C\Big\}$$

satisfies $P_0(A_{n,C}) \geq 1 - \delta$ for all $n \geq n_0$. We carry out the deterministic bounded-class arguments on $A_{n,C}$, where all implicit constants may depend on C but not on n. Since δ is arbitrary, the resulting rates and distributional statements hold unconditionally.

Lemma 3 (Isotonic L^2 rate)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$\|m^\star_{n,\text{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

Proof of Lemma 3. Let $T := m(X)$ and let

$$\mathcal{M}_C := \{f(T) : f \in \mathcal{F}_{\text{iso}},\ \|f(T)\|_\infty \leq C\}.$$

By Assumption 2, $g_0(T)$ is bounded by $C_0 \leq C$, and on $A_{n,C}$ we have $m^\star_{n,\text{iso}} \in \mathcal{M}_C$. Since clipping any monotone candidate to $[-C, C]$ preserves monotonicity and can only decrease squared risk against $g_0(T)$,

the population isotonic projection m_0 also belongs to $\mathcal{M}_C$. Thus the problem is a one-dimensional isotonic least-squares regression of Y on the score T. Let

$$g_0(T) := \mathbb{E}[Y \mid T].$$

Then, for every $m \in \mathcal{M}_C$,

$$P_0\{(Y-m)^2\} = P_0\{(Y-g_0)^2\} + P_0\{(g_0-m)^2\},$$

which shows that m_0 is the $L^2(P_T)$-projection of g_0 onto the closed convex set $\mathcal{M}_C$.

Now define the centered class

$$\mathcal{F}_0 := \{g = m - m_0 : \ m \in \mathcal{M}_C\}.$$

For $g \in \mathcal{F}_0$, let $\ell_g(Z) := (Y - m_0(T) - g(T))^2 - (Y - m_0(T))^2$, where $Z = (X, Y)$. Since $m^\star_{n,\mathrm{iso}}$ minimizes the empirical squared risk over $\mathcal{F}_{\mathrm{iso}}$, on $A_{n,C}$ the random element $\hat{g} := m^\star_{n,\mathrm{iso}} - m_0$ belongs to $\mathcal{F}_0$ and satisfies

$$\mathbb{P}_n^L \ell_{\hat{g}} \le 0 = \mathbb{P}_n^L \ell_0.$$

Moreover,

$$P_0 \ell_g = \|g\|^2_{2,P_{0,X}} - 2P_0\left[(Y - m_0)g\right].$$

Because g is measurable with respect to T, the tower property gives

$$P_0\left[(Y - m_0)g\right] = P_0\left[(g_0 - m_0)g\right].$$

Since m_0 is the $L^2(P_T)$-projection of g_0 onto the convex set $\mathcal{M}_C$, the Hilbert-space projection inequality yields

$$P_0\left[(g_0 - m_0)(m - m_0)\right] \le 0 \qquad \text{for every } m \in \mathcal{M}_C.$$

Hence

$$P_0 \ell_g \ge \|g\|^2_{2,P_{0,X}},$$

so squared loss has the required quadratic margin around m_0.

We next bound the modulus of continuity of the empirical process indexed by the localized classes

$$\mathcal{F}_\delta := \{g \in \mathcal{F}_0 : \|g\|_{2,P_{0,X}} \le \delta\}.$$

Because every $m \in \mathcal{M}_C$ is bounded by C, every $g \in \mathcal{F}_\delta$ is uniformly bounded, and the standard bracketing bound for one-dimensional monotone classes gives

$$\log N_{[]}(\varepsilon, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta/\varepsilon, \qquad 0 < \varepsilon < \delta.$$

Therefore

$$J_{[]}(\delta, \mathcal{F}_\delta, L_2(P_{0,X})) \lesssim \delta^{1/2};$$

see, e.g., van der Vaart and Wellner (1996).

Write $\xi := Y - m_0(T)$, so that $\ell_g = g^2 - 2\xi g$. The class $\{g^2 : g \in \mathcal{F}_\delta\}$ is a bounded Lipschitz image of $\mathcal{F}_\delta$, so by Lemma 3.4.2 of van der Vaart and Wellner (1996) and the same entropy bound,

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}_n^L - P_0)(g^2)| \lesssim \delta^{1/2}.$$

For the multiplier term, Assumption 2 implies that $Y - g_0(T)$ is sub-Gaussian or subexponential. Since both $g_0(T)$ and $m_0(T)$ are bounded by C, the difference $g_0(T) - m_0(T)$ is bounded, and therefore $\xi = Y - m_0(T)$ is also sub-Gaussian or subexponential. Thus the product class $\xi\mathcal{F}_\delta := \{\xi g : g \in \mathcal{F}_\delta\}$ satisfies the same localized bracketing bound in the Bernstein-type norm used in the subexponential least-squares proof of Bibaut and van der Laan (2019), yielding

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}_n^L - P_0)(\xi g)| \lesssim \delta^{1/2}.$$

Combining the previous two displays,

$$E^* \sup_{g \in \mathcal{F}_\delta} \sqrt{n}\,|(\mathbb{P}_n^L - P_0)\ell_g| \lesssim \delta^{1/2}.$$

We may therefore apply Theorem 3.4.1 of van der Vaart and Wellner (1996) with $d(g) = \|g\|_{2,P_{0,X}}$, quadratic margin $P_0\ell_g \gtrsim d^2(g)$, and modulus $\phi_n(\delta) \asymp \delta^{1/2}$. The fixed-point condition $r_n^2\phi_n(r_n^{-1}) \lesssim \sqrt{n}$ gives $r_n \asymp n^{1/3}$, and hence

$$\|m^\star_{n,\mathrm{iso}} - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}), \qquad \|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

The preceding display is obtained on $A_{n,C}$. Since $P_0(A^c_{n,C}) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same rate holds unconditionally. This is the classical one-dimensional isotonic rate, stated directly in the random-design $L^2(P_{0,X})$ norm relevant for the theorem. □

Proof of Lemma 2. Because $h \equiv 1$ belongs to the calibration score class, equation (8) gives

$$\mathbb{P}_n^L\{Y - m^\star_{n,\mathrm{iso}}(X)\} = 0.$$

Hence

$$\mathbb{P}_n^L Y - \mathbb{P}_n^L m_0(X) = \mathbb{P}_n^L\{m^\star_{n,\mathrm{iso}}(X) - m_0(X)\}.$$

By Cauchy–Schwarz and Lemma 3,

$$P_{0,X}\left|m^\star_{n,\mathrm{iso}} - m_0\right| \le \|m^\star_{n,\mathrm{iso}} - m_0\|_{2,P_{0,X}} = O_p(n^{-1/3}),$$

and Lemma 4 gives

$$(\mathbb{P}_n^L - P_{0,X})\{m^\star_{n,\mathrm{iso}} - m_0\} = O_p(n^{-2/3}).$$

Therefore

$$\mathbb{P}_n^L\{m^\star_{n,\mathrm{iso}}(X) - m_0(X)\} = P_{0,X}\{m^\star_{n,\mathrm{iso}} - m_0\} + (\mathbb{P}_n^L - P_{0,X})\{m^\star_{n,\mathrm{iso}} - m_0\} = o_p(1).$$

By Assumption 1, the law of large numbers yields $\mathbb{P}_n^L Y \to_p \psi_0$ and $\mathbb{P}_n^L m_0(X) \to_p \mathbb{E}_{P_{0,X}}[m_0(X)]$. Therefore $\mathbb{E}_{P_{0,X}}[m_0(X)] = \psi_0$. □

Lemma 4 (Centered empirical-process bounds for isotonic calibration)**.** *Suppose Assumptions 1 and 2 holds. Then*

$$(\mathbb{P}_n^L - P_{0,X})\{m^\star_{n,\mathrm{iso}}(X) - m_0(X)\} = O_p(n^{-2/3}), \qquad (\mathbb{P}_N^U - P_{0,X})\{m^\star_{n,\mathrm{iso}}(\widetilde{X}) - m_0(\widetilde{X})\} = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Proof of Lemma 4. By Lemma 3, the isotonic fit satisfies

$$\|m^\star_{n,\mathrm{iso}} - m_0\|^2_{2,P_{0,X}} = O_p(n^{-2/3}).$$

To control the centered empirical-process terms, consider the localized class

$$\mathcal{F}_{n,C} := \{g = f \circ m - m_0 :\ f \in \mathcal{F}_{\mathrm{iso}},\ \|f \circ m\|_\infty \le C,\ \|g\|^2_{2,P_{0,X}} \lesssim n^{-2/3}\}.$$

On $A_{n,C}$, the function $m^\star_{n,\mathrm{iso}} - m_0$ belongs to $\mathcal{F}_{n,C}$. Moreover, $\mathcal{F}_{n,C}$ has a bounded envelope because both $f \circ m$ and m_0 are bounded by C. Because m is fixed, $\mathcal{F}_{n,C}$ is a localized class of bounded monotone transformations of a one-dimensional score, and standard entropy bounds for monotone classes imply a finite entropy integral. Applying a local maximal inequality for empirical processes, such as Lemma 3.4.2 of van der Vaart and Wellner (1996), at localization radius $n^{-1/3}$ then gives, on $A_{n,C}$,

$$|(\mathbb{P}_n^L - P_{0,X})\{m^\star_{n,\mathrm{iso}} - m_0\}| = O_p(n^{-2/3}).$$

Since $P_0(A^c_{n,C}) \le \delta$ for all large n, and $\delta > 0$ was arbitrary, the same bound holds unconditionally. For the unlabeled sample, conditional on the labeled data, $m^\star_{n,\mathrm{iso}} - m_0$ is fixed and the unlabeled sample is independent, so

$$(\mathbb{P}_N^U - P_{0,X})\{m^\star_{n,\mathrm{iso}} - m_0\} = O_p\left(N^{-1/2}\|m^\star_{n,\mathrm{iso}} - m_0\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big)$$

by Lemma 3. □

G Proof of asymptotic linearity

Proof of theorem 2. By Lemma 2, $P_{0,X} m_0 = \psi_0$. Using theorem 1 with $\widehat{w} \equiv 1$,

$$\widehat{\psi}_{\text{iso}} = \rho_n \mathbb{P}_n^L\{m_{n,\text{iso}}^\star(X)\} + (1-\rho_n)\mathbb{P}_N^U\{m_{n,\text{iso}}^\star(\widetilde{X})\} + \rho_n \mathbb{P}_n^L\{Y - m_{n,\text{iso}}^\star(X)\}.$$

Add and subtract the AIPW form built from m_0:

$$\widehat{\psi}_{\text{iso}} = \rho_n \mathbb{P}_n^L\left\{m_0 + \rho_n^{-1}(Y - m_0)\right\} + (1-\rho_n)\mathbb{P}_N^U\{m_0(\widetilde{X})\} + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\text{iso}}^\star - m_0\}.$$

Subtracting ψ_0 gives

$$\begin{aligned}\widehat{\psi}_{\text{iso}} - \psi_0 = \rho_n(\mathbb{P}_n^L - P_0)\Big\{m_0 - \psi_0 + \rho_n^{-1}(Y - m_0)\Big\} \\ + (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})(m_0 - \psi_0) \\ + (1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\text{iso}}^\star - m_0\}.\end{aligned}$$

To replace ρ_n^{-1} by ρ_0^{-1}, note that

$$\rho_n(\mathbb{P}_n^L - P_0)\Big\{(\rho_n^{-1} - \rho_0^{-1})(Y - m_0)\Big\} = \Big(1 - \frac{\rho_n}{\rho_0}\Big)(\mathbb{P}_n^L - P_0)(Y - m_0).$$

The right-hand side is $o_p(M^{-1/2})$ because $\rho_n \to \rho_0$ and $(\mathbb{P}_n^L - P_0)(Y - m_0) = O_p(n^{-1/2}) = O_p(M^{-1/2})$. Also,

$$(1-\rho_n)(\mathbb{P}_N^U - \mathbb{P}_n^L)\{m_{n,\text{iso}}^\star - m_0\} = (1-\rho_n)(\mathbb{P}_N^U - P_{0,X})\{m_{n,\text{iso}}^\star - m_0\} - (1-\rho_n)(\mathbb{P}_n^L - P_{0,X})\{m_{n,\text{iso}}^\star - m_0\},$$

so the labeled term is $O_p(n^{-2/3})$ and the unlabeled term is $O_p(n^{-1/3}N^{-1/2})$ by Lemma 4. Thus the remainder $R_{n,N}$, defined as the difference between $\widehat{\psi}_{\text{iso}} - \psi_0$ and the two leading empirical-process terms in equation (19), satisfies the displayed bound in theorem 2. Since each component is $o_p(M^{-1/2})$, this also yields equation (19). □

Proof of corollary 1. By theorem 2,

$$\sqrt{M}\,(\widehat{\psi}_{\text{iso}} - \psi_0) = \frac{1}{\sqrt{M}}\sum_{i=1}^n D_{m_0}^L(X_i, Y_i) + \frac{1}{\sqrt{M}}\sum_{j=1}^N D_{m_0}^U(\widetilde{X}_j) + o_p(1).$$

The labeled and unlabeled samples are independent, each summand has mean zero, and $\rho_n \to \rho_0 \in (0,1)$ by Assumption 1. A triangular-array central limit theorem therefore gives the stated asymptotic normal law with variance σ_0^2.

For the variance estimator, let $g_n := m_{n,\text{iso}}^\star - m_0$. By Lemma 3,

$$P_{0,X}(g_n^2) = \|g_n\|_{2,P_{0,X}}^2 = O_p(n^{-2/3}).$$

Moreover, the proof of Lemma 3 already established the localized empirical-process bound

$$\sup_{g \in \mathcal{F}_\delta} |(\mathbb{P}_n^L - P_{0,X})(g^2)| = O_p(n^{-1/2}\delta^{1/2}),$$

for the squared class $\{g^2 : g \in \mathcal{F}_\delta\}$. Taking $\delta \asymp n^{-1/3}$, which matches the localization radius of g_n, gives

$$(\mathbb{P}_n^L - P_{0,X})(g_n^2) = O_p(n^{-2/3}).$$

Hence

$$\mathbb{P}_n^L(g_n^2) = O_p(n^{-2/3}) = o_p(1).$$

On the boundedness event $A_{n,C}$, conditional on the labeled data, g_n is fixed and $|g_n| \le 2C$, so

$$(\mathbb{P}_N^U - P_{0,X})(g_n^2) = O_p\left(N^{-1/2}\|g_n^2\|_{2,P_{0,X}}\right) = O_p\left(C\,N^{-1/2}\|g_n\|_{2,P_{0,X}}\right) = O_p\big(n^{-1/3}N^{-1/2}\big).$$

Therefore

$$\mathbb{P}_N^U(g_n^2) = O_p(n^{-2/3}) + O_p\big(n^{-1/3}N^{-1/2}\big) = o_p(1).$$

Since $P_0(A_{n,C}^c) \le \delta$ for all large n and arbitrary $\delta > 0$, these bounds hold unconditionally. Also, $\widehat{\psi}_{\text{iso}} \to_p \psi_0$ follows from theorem 2. Finally,

$$\widehat{D}^L - D_{m_0}^L = (1-\rho_n^{-1})g_n - (\widehat{\psi}_{\text{iso}} - \psi_0) + (\rho_n^{-1} - \rho_0^{-1})(Y - m_0),$$

and

$$\widehat{D}^U - D_{m_0}^U = g_n(\widetilde{X}) - (\widehat{\psi}_{\text{iso}} - \psi_0).$$

Because $\rho_n \to \rho_0$, $Y - m_0$ has finite second moment, and $\mathbb{P}_n^L(g_n^2), \mathbb{P}_N^U(g_n^2) = o_p(1)$, we obtain

$$\widehat{D}_i^L - D_{m_0}^L(X_i, Y_i) = o_p(1) \qquad \text{and} \qquad \widehat{D}_j^U - D_{m_0}^U(\widetilde{X}_j) = o_p(1)$$

in empirical L^2, which yields $\widehat{\sigma}^2 \to_p \sigma_0^2$. The Wald interval claim then follows by Slutsky's theorem. □

H Proof of reduced-model efficiency

Proof of corollary 2. Part (i) follows because, under the conditions of theorem 2, the influence pair in theorem 2 is exactly the pair in proposition 6. Part (ii) follows because if $m_0 = \mu_0$ almost surely, then the same influence pair also coincides with the efficient influence pair in proposition 2. Part (iii) follows because monotonicity of $\mathbb{E}[Y \mid m(X)]$ in $m(X)$ identifies the score-based regression with the isotonic target m_0, so the reduced model indexed by m_0 is the same as the model indexed by the original score $m(X)$. □

I Pooled i.i.d. Missing-Data Formulation

The joint two-sample setup can be embedded into a pooled i.i.d. model by introducing a sample-membership indicator $R \in \{0, 1\}$, where $R = 1$ denotes labeled observations and $R = 0$ denotes unlabeled observations. In that formulation one observes i.i.d. data $O = (X, R, RY)$ with $\mathbb{P}(R = 1) = \rho_0$, and the calibrated estimator becomes

$$\widehat{\psi}_{\text{cal}} = \mathbb{P}_M(m_n^\star) + \mathbb{P}_M\left[\frac{R}{\rho_0}\{Y - m_n^\star\}\right]$$

up to the same calibration identity as in the main text. The corresponding observed-data efficient influence function is

$$m_0(X) - \psi_0 + \frac{R}{\rho_0}\{Y - m_0(X)\},$$

which is the single-sample analogue of the reduced-model influence pair in proposition 6. Thus the usual i.i.d. missing-data formulation is a convenient special case, but the main text keeps the two-dataset structure explicit.

J Additional empirical details

This appendix records the benchmark construction used in section 5.2. We use the official `ppi_py` real-data mean-estimation examples and preserve their labeled-sample-size grids. The `forest` experiment uses $n \in \{50, 100, \ldots, 500\}$, while `galaxies` and semisupervised `census_income` use $n \in \{50, 155, 261, 366, 472, 577, 683, 788, 894, 1000\}$. For each n, the remaining observations form the unlabeled sample. We rerandomize this split 5000 times at each n, always evaluating all estimators on the same split and taking the full-sample mean as the benchmark truth.

The estimator set mirrors the original PPI comparison wherever possible. We report the labeled-only estimator, the imputation benchmark on binary-outcome tasks, the official semisupervised linear baseline from `ppi_py`,

PPI computed using the official `ppi_py` implementation, and the classical AIPW estimator based on the same prediction score. We then add four calibration-based plug-in estimators: linear calibration; Platt scaling on the binary-outcome benchmarks; isotonic calibration with a fixed minimum bin size of 10; and Venn–Abers shrinkage on the binary-outcome benchmarks. Platt scaling and Venn–Abers are omitted on `census_income` because that outcome is not binary. As discussed in the main text, PPI++ and prognostic-score adjustment are not shown as separate baselines because, in this one-dimensional score setting, they lie on the same first-order correction path as linear calibration.

For the calibration-based plug-in estimators, we fit the calibrator on the labeled pairs $\{(m(X_i), Y_i)\}_{i=1}^n$, transform both the labeled and unlabeled scores, and then compute the pooled plug-in mean together with the same semisupervised influence-style Wald interval used throughout the benchmark code. AIPW uses the same score but averages it over the pooled covariate sample before adding the labeled residual correction, whereas PPI uses only the unlabeled score average. In theory this retains less labeled-sample score information, though in the finite-sample benchmark results the empirical variance gap is often small and not uniform across all datasets. Linear calibration regresses Y on an intercept and the score, then clips the fitted values to the range of predicted outcomes observed in the labeled data to prevent extrapolation on the unlabeled data. Platt scaling fits a logistic regression of the binary outcome on a stabilized logit transform of the score. The fixed-bin isotonic estimator also clips to the labeled outcome range. The reported summaries are Monte Carlo averages of bias, empirical variance, MSE, interval coverage, and relative efficiency versus PPI across the repeated random splits. The complete numerical outputs, including dataset-level tables and per-metric plots, are generated automatically by the accompanying reproduction pipeline.

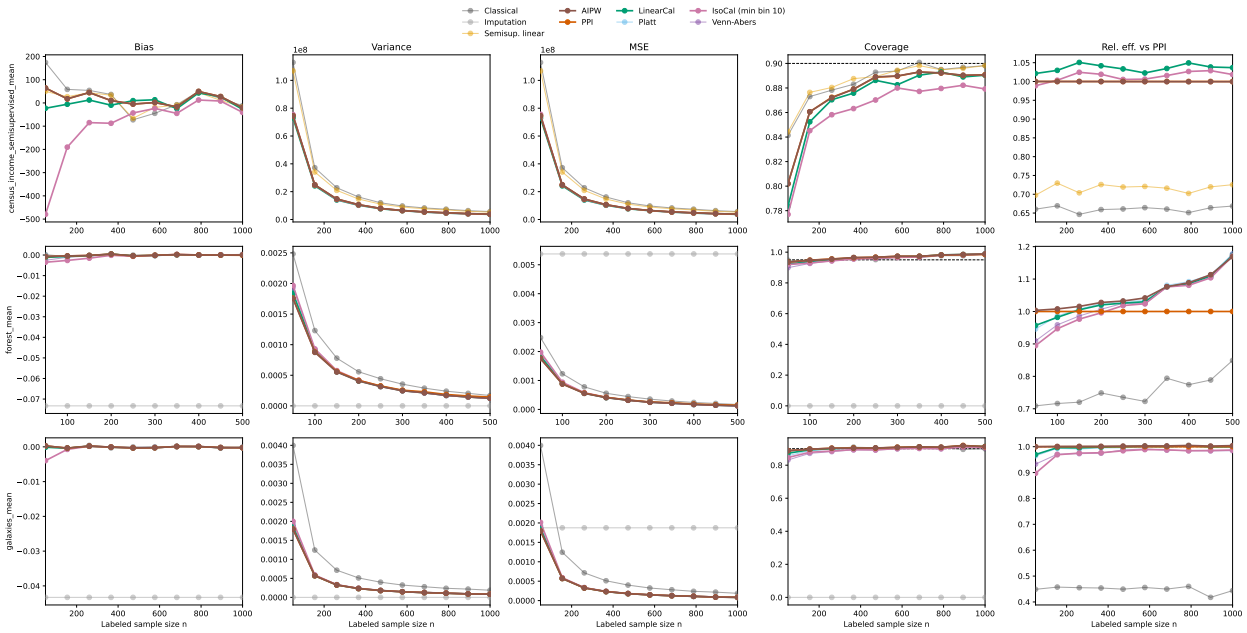

Figure 3: Full diagnostic grid for the reproduced PPI benchmarks, showing bias, empirical variance, MSE, coverage, and relative efficiency versus PPI across labeled-sample-size regimes. The dashed horizontal line in the coverage panels marks the nominal target coverage $1 - \alpha$, and the dashed horizontal line in the relative-efficiency panels marks parity with PPI. Platt scaling and Venn–Abers appear only on the binary-outcome datasets where they are well defined.

K Python Code

K.1 Isotonic regression via XGBoost

```
import numpy as np
import xgboost as xgb

def isoreg_with_xgboost(x, y, max_depth=15, min_child_weight=20, weights=None):
    """
    Fit a monotone calibrator f so that f(x) is nondecreasing in x.

    Parameters
    ----------
    x : array-like, shape (n,) or (n, 1)
        Predictor used for calibration. In the PPI application this is usually
        the score m(X), so x is typically one-dimensional.
    y : array-like, shape (n,)
        Labeled outcomes.
    max_depth : int, default=15
        Maximum depth for the one-round XGBoost fit.
    min_child_weight : float, default=20
        Minimum total weight in a terminal node.
    weights : array-like, optional
        Optional observation weights.

    Returns
    -------
    predict_fn : callable
        Function mapping new x values to calibrated predictions f(x).
    """
    x = np.asarray(x)
    y = np.asarray(y).reshape(-1)
    x = x.reshape(len(y), -1)

    data = xgb.DMatrix(data=x, label=y, weight=weights)
    params = {
        "max_depth": max_depth,
        "min_child_weight": min_child_weight,
        "monotone_constraints": "(" + ",".join(["1"] * x.shape[1]) + ")",
        "eta": 1.0,
        "gamma": 0.0,
        "lambda": 0.0,
        "objective": "reg:squarederror",
        "verbosity": 0,
    }
    iso_fit = xgb.train(params=params, dtrain=data, num_boost_round=1)

    def predict_fn(x_new):
        x_new = np.asarray(x_new)
        if x_new.ndim == 1:
            x_new = x_new.reshape(-1, 1)
        data_pred = xgb.DMatrix(data=x_new)
        return iso_fit.predict(data_pred)

    return predict_fn
```

K.2 Isotonic-calibrated plug-in

```
import numpy as np

from statistics import NormalDist


def _as_1d_array(x, name):
    x = np.asarray(x).reshape(-1)
    if x.size == 0:
        raise ValueError(f"{name} must be non-empty.")
    return x


def _resolve_scores(
    X_labeled=None,
    Y_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    """
    Resolve the initial score m(X) on the labeled and unlabeled samples.

    Users may either supply:
    1. a scoring function `score_fn`, together with `X_labeled` and
       `X_unlabeled`, or
    2. precomputed scores `score_labeled` and `score_unlabeled`.
    """
    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn with X_labeled/X_unlabeled, or provide "
            "both score_labeled and score_unlabeled."
        )

    score_labeled = _as_1d_array(score_labeled, "score_labeled")
    score_unlabeled = _as_1d_array(score_unlabeled, "score_unlabeled")

    if Y_labeled is not None and len(score_labeled) != len(Y_labeled):
        raise ValueError("score_labeled and Y_labeled must have the same length.")

    return score_labeled, score_unlabeled


def calibrated_plugin_ppi(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
```

```
    min_obs=20,
    alpha=0.05,
    inference="wald",
    n_bootstrap=999,
    random_state=None,
    max_depth=15,
):
    """
    Calibrated plug-in PPI estimator for psi = E[Y].

    The estimator first calibrates the initial score m(X) on the labeled data,
    then averages the calibrated score over the pooled labeled and unlabeled
    covariate sample.

    Parameters
    ----------
    Y_labeled : array-like, shape (n,)
        Outcomes in the labeled sample.
    X_labeled : array-like, optional
        Covariates in the labeled sample. Needed only when score_fn is used.
    X_unlabeled : array-like, optional
        Covariates in the unlabeled sample. Needed only when score_fn is used.
    score_fn : callable, optional
        Function returning the initial score m(X).
    score_labeled : array-like, optional
        Precomputed values m(X) on the labeled sample.
    score_unlabeled : array-like, optional
        Precomputed values m(Xtilde) on the unlabeled sample.
    min_obs : int, default=20
        Minimum node size used in the one-round monotone XGBoost calibrator.
    alpha : float, default=0.05
        Miscoverage level for confidence intervals.
    inference : {"wald", "bootstrap"}, default="wald"
        Type of confidence interval to report.
    n_bootstrap : int, default=999
        Number of bootstrap draws when inference="bootstrap".
    random_state : int, optional
        Seed for the bootstrap.
    max_depth : int, default=15
        Maximum depth for the calibration fit.

    Returns
    -------
    result : dict
        Dictionary containing the estimate, standard error, confidence
        interval, and the calibrated scores on both samples.

    Notes
    -----
    This implementation treats the initial score m(X) as fixed. If the score
    is itself estimated on the labeled data, cross-fitting should be used.
    """
    Y_labeled = _as_1d_array(Y_labeled, "Y_labeled")
    score_labeled, score_unlabeled = _resolve_scores(
        X_labeled=X_labeled,
        Y_labeled=Y_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
```

```
)

n = len(Y_labeled)
N = len(score_unlabeled)
M = n + N
rho_hat = n / M

calibrator = isoreg_with_xgboost(
    x=score_labeled,
    y=Y_labeled,
    max_depth=max_depth,
    min_child_weight=min_obs,
)

calibrated_labeled = calibrator(score_labeled)
calibrated_unlabeled = calibrator(score_unlabeled)

psi_hat = (
    n * calibrated_labeled.mean() + N * calibrated_unlabeled.mean()
) / M

# Estimated influence-function contributions for Wald inference.
if_labeled = calibrated_labeled - psi_hat + (Y_labeled - calibrated_labeled) / rho_hat
if_unlabeled = calibrated_unlabeled - psi_hat
asymptotic_var = rho_hat * np.var(if_labeled, ddof=1) + (1 - rho_hat) * np.var(
    if_unlabeled, ddof=1
)
se_wald = np.sqrt(asymptotic_var / M)

z = NormalDist().inv_cdf(1 - alpha / 2)
ci_wald = (psi_hat - z * se_wald, psi_hat + z * se_wald)

result = {
    "estimate": psi_hat,
    "standard_error": se_wald,
    "confidence_interval": ci_wald,
    "inference": "wald",
    "alpha": alpha,
    "rho_hat": rho_hat,
    "calibrator": calibrator,
    "score_labeled": score_labeled,
    "score_unlabeled": score_unlabeled,
    "calibrated_labeled": calibrated_labeled,
    "calibrated_unlabeled": calibrated_unlabeled,
}

if inference == "wald":
    return result

if inference != "bootstrap":
    raise ValueError("inference must be either 'wald' or 'bootstrap'.")

rng = np.random.default_rng(random_state)
bootstrap_estimates = np.empty(n_bootstrap)

for b in range(n_bootstrap):
    idx_l = rng.integers(0, n, size=n)
    idx_u = rng.integers(0, N, size=N)

    calibrator_b = isoreg_with_xgboost(
```

```
            x=score_labeled[idx_l],
            y=Y_labeled[idx_l],
            max_depth=max_depth,
            min_child_weight=min_obs,
        )
        calibrated_labeled_b = calibrator_b(score_labeled[idx_l])
        calibrated_unlabeled_b = calibrator_b(score_unlabeled[idx_u])

        bootstrap_estimates[b] = (
            n * calibrated_labeled_b.mean() + N * calibrated_unlabeled_b.mean()
        ) / M

    ci_boot = tuple(
        np.quantile(bootstrap_estimates, [alpha / 2, 1 - alpha / 2])
    )
    result.update(
        {
            "standard_error": np.std(bootstrap_estimates, ddof=1),
            "confidence_interval": ci_boot,
            "inference": "bootstrap",
            "bootstrap_estimates": bootstrap_estimates,
        }
    )
    return result


# Example 1: supply a score function m(X)
# fit = calibrated_plugin_ppi(
#     Y_labeled=Y,
#     X_labeled=X,
#     X_unlabeled=X_tilde,
#     score_fn=m,
#     min_obs=20,
#     inference="wald",
# )

# Example 2: supply precomputed scores m(X) and m(Xtilde)
# fit = calibrated_plugin_ppi(
#     Y_labeled=Y,
#     score_labeled=mX,
#     score_unlabeled=mX_tilde,
#     min_obs=20,
#     inference="bootstrap",
#     n_bootstrap=999,
#     random_state=1,
# )
```

K.3 Linear calibration

```
import numpy as np

from statistics import NormalDist


def _as_1d_array(x, name):
    x = np.asarray(x).reshape(-1)
    if x.size == 0:
        raise ValueError(f"{name} must be non-empty.")
    return x
```

```
def _resolve_scores(
    X_labeled=None,
    Y_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
):
    if score_fn is not None:
        if X_labeled is None or X_unlabeled is None:
            raise ValueError(
                "If score_fn is provided, then X_labeled and X_unlabeled "
                "must also be provided."
            )
        score_labeled = score_fn(X_labeled)
        score_unlabeled = score_fn(X_unlabeled)
    elif score_labeled is None or score_unlabeled is None:
        raise ValueError(
            "Provide either score_fn with X_labeled/X_unlabeled, or provide "
            "both score_labeled and score_unlabeled."
        )

    score_labeled = _as_1d_array(score_labeled, "score_labeled")
    score_unlabeled = _as_1d_array(score_unlabeled, "score_unlabeled")

    if Y_labeled is not None and len(score_labeled) != len(Y_labeled):
        raise ValueError("score_labeled and Y_labeled must have the same length.")

    return score_labeled, score_unlabeled


def linear_calibrated_plugin_ppi(
    Y_labeled,
    X_labeled=None,
    X_unlabeled=None,
    score_fn=None,
    score_labeled=None,
    score_unlabeled=None,
    alpha=0.05,
):
    """
    Linear-calibrated plug-in PPI estimator for psi = E[Y].

    The calibration model is
        E[Y | m(X)] approx beta_0 + beta_1 m(X),
    estimated by least squares on the labeled sample. To avoid extrapolation,
    scores are clipped to the labeled score range before applying the
    calibrator to either sample.
    """
    Y_labeled = _as_1d_array(Y_labeled, "Y_labeled")
    score_labeled, score_unlabeled = _resolve_scores(
        X_labeled=X_labeled,
        Y_labeled=Y_labeled,
        X_unlabeled=X_unlabeled,
        score_fn=score_fn,
        score_labeled=score_labeled,
        score_unlabeled=score_unlabeled,
    )
```

```
    n = len(Y_labeled)
    N = len(score_unlabeled)
    M = n + N
    rho_hat = n / M

    design = np.column_stack([np.ones(n), score_labeled])
    beta_hat, _, _, _ = np.linalg.lstsq(design, Y_labeled, rcond=None)
    score_range = (score_labeled.min(), score_labeled.max())

    def calibrator(score):
        score = np.asarray(score).reshape(-1)
        score = np.clip(score, score_range[0], score_range[1])
        return beta_hat[0] + beta_hat[1] * score

    calibrated_labeled = calibrator(score_labeled)
    calibrated_unlabeled = calibrator(score_unlabeled)

    psi_hat = (
        n * calibrated_labeled.mean() + N * calibrated_unlabeled.mean()
    ) / M

    if_labeled = calibrated_labeled - psi_hat + (Y_labeled - calibrated_labeled) / rho_hat
    if_unlabeled = calibrated_unlabeled - psi_hat
    asymptotic_var = rho_hat * np.var(if_labeled, ddof=1) + (1 - rho_hat) * np.var(
        if_unlabeled, ddof=1
    )
    standard_error = np.sqrt(asymptotic_var / M)

    z = NormalDist().inv_cdf(1 - alpha / 2)
    confidence_interval = (
        psi_hat - z * standard_error,
        psi_hat + z * standard_error,
    )

    return {
        "estimate": psi_hat,
        "standard_error": standard_error,
        "confidence_interval": confidence_interval,
        "alpha": alpha,
        "rho_hat": rho_hat,
        "beta_hat": beta_hat,
        "score_range": score_range,
        "score_labeled": score_labeled,
        "score_unlabeled": score_unlabeled,
        "calibrated_labeled": calibrated_labeled,
        "calibrated_unlabeled": calibrated_unlabeled,
    }


# Example: linear calibration with a score function m(X)
# fit = linear_calibrated_plugin_ppi(
#     Y_labeled=Y,
#     X_labeled=X,
#     X_unlabeled=X_tilde,
#     score_fn=m,
# )
```

Dataset	n	N	Estimator	Bias	Variance	MSE	Coverage	RelEff
census_income_semisupervised_mean	50	380041	AIPW	64.1415	74530505.1597	74534619.2853	0.802	1.000
census_income_semisupervised_mean	50	380041	Classical	174.3296	112879690.3397	112910081.1357	0.841	0.660
census_income_semisupervised_mean	50	380041	IsoCal (min bin 10)	-479.6340	75379435.8991	75609484.6737	0.777	0.989
census_income_semisupervised_mean	50	380041	LinearCal	-22.9713	72975497.5814	72976025.2610	0.784	1.021
census_income_semisupervised_mean	50	380041	PPI	64.1270	74528348.5082	74532460.7743	0.802	1.000
census_income_semisupervised_mean	50	380041	Semisup. linear	49.6769	106851003.3081	106853471.1059	0.844	0.697
census_income_semisupervised_mean	577	379514	AIPW	1.6631	6451762.8714	6451765.6373	0.890	1.000
census_income_semisupervised_mean	577	379514	Classical	-44.6870	9708740.7725	9710737.7047	0.894	0.664
census_income_semisupervised_mean	577	379514	IsoCal (min bin 10)	-23.7251	6409823.1141	6410385.9947	0.880	1.006
census_income_semisupervised_mean	577	379514	LinearCal	13.6244	6307041.6066	6307227.2310	0.883	1.023
census_income_semisupervised_mean	577	379514	PPI	1.7336	6449769.3295	6449772.3348	0.890	1.000
census_income_semisupervised_mean	577	379514	Semisup. linear	-19.0374	8948168.5056	8948530.9279	0.894	0.721
census_income_semisupervised_mean	1000	379091	AIPW	-20.5346	3817516.0488	3817937.7195	0.891	0.999
census_income_semisupervised_mean	1000	379091	Classical	-11.5142	5706667.9025	5706800.4785	0.898	0.669
census_income_semisupervised_mean	1000	379091	IsoCal (min bin 10)	-40.5935	3746139.4177	3747787.2512	0.879	1.018
census_income_semisupervised_mean	1000	379091	LinearCal	-27.7615	3679849.2001	3680619.8996	0.890	1.037
census_income_semisupervised_mean	1000	379091	PPI	-20.5584	3815435.7633	3815858.4117	0.891	1.000
census_income_semisupervised_mean	1000	379091	Semisup. linear	-17.5832	5259398.9756	5259708.1447	0.898	0.725
forest_mean	50	1546	AIPW	-0.0007	0.0018	0.0018	0.934	1.003
forest_mean	50	1546	Classical	0.0002	0.0025	0.0025	0.948	0.709
forest_mean	50	1546	Imputation	-0.0733	0.0000	0.0054	0.000	inf
forest_mean	50	1546	IsoCal (min bin 10)	-0.0035	0.0020	0.0020	0.919	0.895
forest_mean	50	1546	LinearCal	-0.0012	0.0018	0.0018	0.922	0.957
forest_mean	50	1546	Platt	-0.0002	0.0019	0.0019	0.915	0.946
forest_mean	50	1546	PPI	-0.0007	0.0018	0.0018	0.935	1.000
forest_mean	50	1546	Venn-Abers	-0.0026	0.0019	0.0019	0.899	0.909
forest_mean	300	1296	AIPW	-0.0001	0.0002	0.0002	0.974	1.042
forest_mean	300	1296	Classical	-0.0003	0.0004	0.0004	0.967	0.723
forest_mean	300	1296	Imputation	-0.0733	0.0000	0.0054	0.000	inf
forest_mean	300	1296	IsoCal (min bin 10)	-0.0002	0.0003	0.0003	0.966	1.024
forest_mean	300	1296	LinearCal	-0.0002	0.0002	0.0002	0.971	1.030
forest_mean	300	1296	Platt	-0.0001	0.0002	0.0002	0.971	1.033
forest_mean	300	1296	PPI	-0.0001	0.0003	0.0003	0.973	1.000
forest_mean	300	1296	Venn-Abers	-0.0002	0.0003	0.0003	0.966	1.024
forest_mean	500	1096	AIPW	0.0000	0.0001	0.0001	0.988	1.169
forest_mean	500	1096	Classical	0.0003	0.0002	0.0002	0.984	0.849
forest_mean	500	1096	Imputation	-0.0733	0.0000	0.0054	0.000	inf
forest_mean	500	1096	IsoCal (min bin 10)	0.0001	0.0001	0.0001	0.987	1.171
forest_mean	500	1096	LinearCal	0.0000	0.0001	0.0001	0.988	1.173
forest_mean	500	1096	Platt	0.0000	0.0001	0.0001	0.988	1.177
forest_mean	500	1096	PPI	-0.0001	0.0001	0.0001	0.985	1.000
forest_mean	500	1096	Venn-Abers	0.0001	0.0001	0.0001	0.986	1.177
galaxies_mean	50	16693	AIPW	0.0003	0.0018	0.0018	0.888	1.000
galaxies_mean	50	16693	Classical	0.0004	0.0040	0.0040	0.883	0.449
galaxies_mean	50	16693	Imputation	-0.0433	0.0000	0.0019	0.000	inf
galaxies_mean	50	16693	IsoCal (min bin 10)	-0.0040	0.0020	0.0020	0.847	0.897
galaxies_mean	50	16693	LinearCal	-0.0001	0.0019	0.0019	0.872	0.970
galaxies_mean	50	16693	Platt	0.0002	0.0019	0.0019	0.858	0.965
galaxies_mean	50	16693	PPI	0.0003	0.0018	0.0018	0.888	1.000
galaxies_mean	50	16693	Venn-Abers	-0.0001	0.0019	0.0019	0.832	0.933
galaxies_mean	577	16166	AIPW	-0.0002	0.0001	0.0001	0.908	1.003
galaxies_mean	577	16166	Classical	-0.0000	0.0003	0.0003	0.910	0.456
galaxies_mean	577	16166	Imputation	-0.0433	0.0000	0.0019	0.000	inf
galaxies_mean	577	16166	IsoCal (min bin 10)	-0.0002	0.0001	0.0001	0.900	0.989
galaxies_mean	577	16166	LinearCal	-0.0002	0.0001	0.0001	0.906	1.001
galaxies_mean	577	16166	Platt	-0.0003	0.0001	0.0001	0.905	0.999
galaxies_mean	577	16166	PPI	-0.0003	0.0001	0.0001	0.908	1.000
galaxies_mean	577	16166	Venn-Abers	-0.0002	0.0001	0.0001	0.899	0.991
galaxies_mean	1000	15743	AIPW	-0.0002	0.0001	0.0001	0.913	1.004
galaxies_mean	1000	15743	Classical	-0.0001	0.0002	0.0002	0.899	0.444
galaxies_mean	1000	15743	Imputation	-0.0433	0.0000	0.0019	0.000	inf
galaxies_mean	1000	15743	IsoCal (min bin 10)	-0.0002	0.0001	0.0001	0.907	0.987
galaxies_mean	1000	15743	LinearCal	-0.0002	0.0001	0.0001	0.913	1.000
galaxies_mean	1000	15743	Platt	-0.0002	0.0001	0.0001	0.910	0.996
galaxies_mean	1000	15743	PPI	-0.0002	0.0001	0.0001	0.912	1.000
galaxies_mean	1000	15743	Venn-Abers	-0.0002	0.0001	0.0001	0.905	0.984

Table 2: Representative numerical summary at the smallest, middle, and largest labeled sample sizes for each reproduced benchmark. The reported quantities are Monte Carlo bias, empirical variance of the point estimator, MSE, Wald-interval coverage, and relative efficiency versus PPI.